

Benchmarked machine learning for neuronal α -synuclein disease integrated staging system (NSD-ISS) prediction with calibrated uncertainty in Parkinson’s disease

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Abstract

Background. The Neuronal α -Synuclein Disease Integrated Staging System (NSD-ISS) redefines Parkinson’s disease (PD) through two biological anchors (S: α -synuclein seed amplification assay; D: dopamine-transporter SPECT), but no computational model predicts NSD-ISS stage from routinely collected clinical data with calibrated uncertainty. **Methods.** Using the Parkinson’s Progression Markers Initiative (PPMI; n=2,201), we benchmark a strict-circularity specification excluding the four variables entering the NSD-ISS staging algorithm, retaining 21 non-staging features across seven clinical domains. Five state-of-the-art tabular methods (CatBoost default+HPO, LightGBM HPO, TabPFN v2, AutoGluon) are evaluated alongside graph-attention baselines, with cross-conformal CV+ uncertainty quantification at three confidence levels. External validation uses BioFIND (n=103) staged by Russo et al. 2025 and the externally-validated Venuto et al. 2025 SAA prediction model for construct-validity assessment of D-anchor-driven labels. **Results.** CatBoost (default) achieves binary NSD+ AUC 0.901 and NSD+ sub-staging AUC 0.908; the five SOTA methods are statistically equivalent at $\epsilon=0.02$ (paired-bootstrap TOST: 34/40 pairs). Removing imaging and genetics features cuts binary AUC by -17.6 percentage points but leaves NSD+ sub-staging unchanged ($\Delta = +0.008$, $p=0.21$), validating a two-stage deployment. Cross-conformal CV+ achieves $>90\%$ internal coverage; external coverage on BioFIND holds for binary (0.915) but collapses for multiclass without recalibration, restored to 0.944 by Mondrian per-class CP on $n \geq 50$ labelled local patients. The Venuto-imputed S+ rate among 647 SAA-unttested NSD-positive PD patients is 91.5%, in the literature band of 88–93%, supporting construct validity of D-anchor-driven labels. **Conclusions.** A strict-circularity NSD-ISS staging benchmark with calibrated uncertainty supports a deployment-ready two-stage clinical pipeline: imaging-anchored binary detection followed by clinical-only sub-staging, with on-site Mondrian conformal recalibration on small labelled local cohorts for multiclass external transport.

Author summary

Parkinson’s disease is now redefined biologically by two markers: α -synuclein protein aggregates detected in cerebrospinal fluid, and dopamine transporter activity measured by brain imaging. Together, they define a six-stage staging system that maps biological progression—but predicting which stage a patient is in, from routine clinical data, has not been computationally benchmarked. We trained machine learning models on 2,201

PPMI patients to predict NSD-ISS stage and quantified uncertainty using conformal prediction. We found that gradient-boosted trees and modern tabular foundation models achieve statistically equivalent performance ($>90\%$ AUC for binary detection), that biological-marker imaging is essential for the initial binary detection but clinical features alone suffice for finer sub-staging, and that distribution-free uncertainty guarantees transport to a new cohort only after on-site recalibration on $\sim 40\text{--}50$ labelled local patients. The work closes the methodology gap for deploying NSD-ISS in clinical practice—particularly at community sites without imaging access—and provides the first construct-validity check using an externally-validated S-anchor predictor to verify that PPMI’s labels are biologically concordant rather than rule-recapitulation artifacts.

Introduction

Parkinson’s disease (PD) has traditionally been staged using clinical severity scales such as the Movement Disorder Society-sponsored Unified Parkinson’s Disease Rating Scale (MDS-UPDRS) [19] and the MDS clinical diagnostic criteria [28]. These scales describe downstream motor manifestations of pathology that has typically been developing for years: pathological staging of Lewy pathology was proposed by Braak *et al.* [6], long before cerebrospinal-fluid alpha-synuclein seed amplification assays and dopamine-transporter (DaT) SPECT made *in vivo* biological staging operational at scale in cohorts such as the Parkinson’s Progression Markers Initiative (PPMI) [24]. In January 2024, Simuni *et al.* [38] proposed the Neuronal alpha-Synuclein Disease Integrated Staging System (NSD-ISS), redefining PD through two biological anchors: neuronal alpha-synuclein (S, assessed by seed amplification assay, SAA) and dopaminergic dysfunction (D, assessed by DaT-SPECT imaging). The NSD-ISS assigns patients to stages 0 through 6, creating a biologically-grounded framework spanning preclinical through severe disability, and has been validated longitudinally: biological stage predicts time-to-clinical-milestones [39], staging replicates across cohorts including BioFIND [32], and 5-year stage transitions show consistent median times of 1.2 years (2B to 3) and 5.0 years (3 to 4) [40].

The NSD-ISS framework has also received important methodological critiques. Espay *et al.* [15] showed that dopaminergic medication status confounds several of the clinical sub-staging variables. They recommended that staging models be trained on cohorts with explicit medication-status stratification—a constraint we adopt in the present work and discuss further in the Discussion.

Despite rapid clinical adoption, the NSD-ISS literature remains predominantly descriptive. While machine-learning approaches have been applied to PD progression and Hoehn–Yahr staging—e.g., Severson *et al.* [63] (LSTM-based PD progression states from MDS-UPDRS), AdaMedGraph [1] (graph propagation for PD clinical milestones on PPMI), Rissanen *et al.* [30] (PD subtype clustering), and Diaz-Rincón *et al.* [12] (conformal prediction for PD medication needs)—none of these approaches specifically targets NSD-ISS biological-stage prediction from routinely-collected PPMI data with calibrated uncertainty. Reporting guidance for such clinical prediction models is provided by the TRIPOD+AI statement [10]. The primary research question of this study is therefore: *can a benchmarked machine-learning framework with calibrated uncertainty predict NSD-ISS biological stage from routinely-collected PPMI clinical and imaging data, including cross-cohort transportability and a clinical-only feature subset for sites lacking DaT-SPECT or SAA access?* Five specific objectives operationalise this question:

1. Benchmark seven tabular and one graph-based model for NSD-ISS stage prediction across four target formulations using PPMI data ($n=2,201$, the Methods).
2. Compare tabular (gradient-boosted trees) versus graph-based (graph attention network (GAT) with patient-similarity graphs) approaches on an identical feature set.
3. Quantify the contribution of biological markers versus clinical-only features, using a feature-ablation protocol matched to site-level resource availability.
4. Evaluate model transportability through external validation on BioFIND ($n=118$, with NSD-ISS ground truth) and prediction-only application to PDBP ($n=893$), using a pre-specified handling protocol for the training-label confound introduced by healthy-control inclusion in PPMI.
5. Calibrate model uncertainty through cross-conformal prediction (CV+) with distribution-free coverage guarantees, so that predictions are reported as small sets of plausible labels when individual-patient confidence is low.

The four NSD-ISS target formulations used throughout this paper span the full clinical-utility spectrum—binary detection of biological disease (screening), three-class triage (treatment routing), full ordinal staging (longitudinal monitoring), and NSD-positive sub-staging (trial enrichment at sites without imaging)—and are summarised in Table 1.

Table 1. Four NSD-ISS target formulations evaluated in this study.

Target	K	What it predicts
Binary	2	NSD+ (stages 1+) vs. NSD− (stage 0)
Three-class	3	Early (0–1) vs. mild (2B) vs. impaired (3–4)
Full ordinal	5	All stages: 0, 1, 2B, 3, 4
NSD-positive	4	NSD+ sub-staging: 1, 2B, 3, 4

Related Work

Machine learning for Parkinson’s disease has focused primarily on clinical outcome prediction and diagnostic classification rather than biological staging. AdaMedGraph [1] combined APPNP graph propagation with AdaBoost (SAMME) for PD progression prediction using PPMI data. We reproduce AdaMedGraph in our benchmark (Table 3 (“Graph-based” rows)) and report its head-to-head performance on the NSD-ISS biological-staging targets; the per-feature-graph ensemble is competitive on binary classification but underperforms tree baselines on multi-class targets, consistent with the broader Grinsztajn 2022 finding. Recent work by Diaz-Rincon *et al.* [12] applied conformal prediction to PD medication needs assessment, establishing distribution-free coverage guarantees in a clinical PD context. However, neither of these approaches—nor any published model to date—targets the NSD-ISS biological staging framework introduced by Simuni *et al.* [38].

The choice of model architecture for clinical prediction tasks remains an active area of investigation. The dominant narrative, established by Grinsztajn *et al.* [21] and Shwartz-Ziv and Armon [37], is that gradient-boosted tree ensembles outperform deep learning on medium-sized tabular datasets. However, more recent work suggests this gap closes at smaller sample sizes: Zabërgja *et al.* [46] report deep-learning methods winning 31 of 34 benchmark tasks at $n < 5,000$, and the closer-look evaluation of Ye *et al.* [45] reaches the same conclusion for modern attention- and transformer-based tabular architectures [20]. Foundation-model tabular approaches have sharpened this shift: Hollmann *et al.* [49] demonstrate that TabPFN v2—a context-based transformer trained on synthetic tabular data—exceeds tuned CatBoost by an average +0.13 AUC across 150 benchmark datasets without any hyperparameter tuning. AutoML ensembling provides a complementary route: AutoGluon [50] achieves consistent leaderboard wins through weighted stacking across a heterogeneous model pool. These developments motivate our four-way tabular-SOTA comparison (CatBoost, LightGBM, TabPFN, AutoGluon) in the Results, which tests whether any single architecture dominates on NSD-ISS biological staging or whether—as Zabërgja, Hollmann, and Erickson collectively predict—tree boosters, foundation-model tabular, and AutoML ensembling converge at the $n \approx 2,000$ clinical-cohort regime where NSD-ISS is defined.

Conformal prediction [43] has emerged as a principled framework for uncertainty quantification in clinical ML, providing finite-sample marginal coverage guarantees without distributional assumptions. Huang *et al.* [22] extended conformal methods to graph neural networks, demonstrating that graph structure can be exploited for tighter prediction sets. Our work applies cross-conformal (CV+) prediction to NSD-ISS staging, combining the distribution-free guarantees of conformal inference with the clinical interpretability of set-valued predictions.

Materials and methods

Study Design

This is a retrospective prediction-model development and external-evaluation study using data from four observational PD cohorts, reported following the TRIPOD+AI guidelines [10]. The methodological pipeline proceeds as: cohort assembly → NSD-ISS staging → feature engineering with strict-circularity exclusion of the four staging-algorithm variables (putamen SBR, MDS-UPDRS-III total, NP1-COG, caudate/putamen ratio) → model training under 5-fold stratified CV → cross-conformal calibration → external validation on BioFIND. The per-cohort participant flow—enrollment, staging eligibility, and final role assignment—is

Fig 1. Study participant flow across the four PD cohorts. PPMI served as the primary training cohort with NSD-ISS ground truth computed from SAA and DaT-SPECT biomarkers. BioFIND provided external validation with NSD-ISS staging replicated from Russo *et al.* [32]. PDBP and HBS were prediction-only cohorts without biological staging data.

summarised in Fig. 1.

Sample size and ethics The PPMI training cohort ($n=2,201$ with 783 binary NSD+ events) exceeds the Riley *et al.* [64] minimum-sample-size criteria for clinical prediction-model development across all four target formulations (events-per-predictor ≥ 19 for NSD+ sub-staging, the lowest-EPV target). PPMI is approved by the Institutional Review Boards of all 33 participating sites and is registered as ClinicalTrials.gov NCT01141023 [24]; BioFIND [32] and the AMP-PD cohorts [14] carry their own per-site IRB approvals. This secondary analysis of fully de-identified data was determined to be exempt from further review by the University of North Dakota Institutional Review Board. Data access was granted under the PPMI Data Use Agreement (PPMI account active since 2024) and AMP-PD Tier-1 DUA (active since 2025).

Data Sources

PPMI ($n=2,201$): A prospective longitudinal study enrolling de novo PD patients, prodromal PD, and healthy controls across 33 sites. AMP-PD v4 data release.

BioFIND ($n=118$ PD): Cross-sectional biomarker study, moderate-to-advanced PD (diagnosed >5 years), 16 US sites. SAA data from three independent laboratories via LONI IDA.

PDBP ($n=893$ PD): NIH-funded longitudinal biomarker cohort. AMP-PD v4 clinical data.

HBS ($n=649$ PD): Single-center longitudinal study at Massachusetts General Hospital. AMP-PD v4 clinical data.

NSD-ISS Biological Staging

NSD-ISS stages were computed from three anchors:

S anchor (synuclein): Seed amplification assay (SAA) positivity. Coverage: 12.6% of PPMI (277/2,201).

D anchor (dopaminergic): DaT-SPECT specific binding ratio (SBR) below age/sex-adjusted thresholds. Coverage: 97.1% (2,137/2,201).

Clinical staging: Among S+ and/or D+ patients, functional impairment assessed via MDS-UPDRS Part II (ADL) and MoCA (cognitive). Stage assignments followed Simuni *et al.* [38].

Feature Engineering

Feature-set hierarchy Paper 1’s primary feature specification is the strict-circularity 21-feature set, chosen so that no variable entering the Simuni 2024 NSD-ISS staging algorithm (either as a direct threshold input or as a component of a derivative feature) remains in the predictor vector. The 21 features span seven clinical domains (Table 2). For external validation where not all features are available across cohorts, we use a 12-feature common subset (AGE_AT_BASELINE, SEX, UPDRS1_TOTAL, UPDRS2_TOTAL, UPDRS3_TREMOR, UPDRS3_RIGIDITY, UPDRS3_BRADYKINESIA, UPDRS3_AXIAL, UPDRS4_TOTAL, MOCA_TOTAL, ESS_TOTAL, RBD_TOTAL), which is the intersection of features present in all four cohorts (PPMI, BioFIND, PDBP, HBS). A pre-registered 33-feature sensitivity variant adding 11 further observed PPMI features (6 cortical thickness, 4 CSF biomarkers, 1 polygenic risk score) is evaluated in two ways: against the Multimodal GAT to test whether the GAT-vs-tree gap is architectural or feature-count-limited (Supplementary S-3, 3-modality 32-feature variant), and against the tabular models to test whether the canonical schema is under-powered (Supplementary S-4, pre-registered null result—the extension does not improve tabular performance and the halt rule fired). Table 3 reports the 22-feature reference specification alongside the 21-feature primary (the two specifications differ only in the inclusion/exclusion of the caudate/putamen ratio; all other features are shared); the strict-exclusion specification is the primary throughout the Results and the Discussion.

Feature retention rationale The 21-feature set retains one score per clinical domain (e.g., SCOPA-AUT total rather than the five subscales), dropping redundant subscale decompositions that share $\geq 80\%$ variance with their totals. Caudate-based DaT-SPECT features are retained (left, right, mean, and asymmetry) because the caudate is a prognostically informative and staging-algorithm-independent striatal sub-region; the putamen SBR and any derivative that depends on it are excluded per the circularity audit below.

Circularity audit (five exclusions) The NSD-ISS framework is itself rule-based: every patient’s stage label is produced by thresholding a specific set of clinical and imaging variables (Simuni *et al.* [38]). A predictor model trained on those same thresholding variables would recover the staging rule rather than extract biological signal from them. To avoid this tautology we exclude five features from the 21-feature primary set:

- **PUTAMEN_L_SBR and PUTAMEN_R_SBR** enter the D-anchor branch of the NSD-ISS staging algorithm directly (dopaminergic deficit is defined as putamen SBR below a population z-score threshold).
- **CAUDATE_PUTAMEN_RATIO** was initially retained on the reasoning that a ratio, not the putamen value itself, carries the biological information; however a pre-registered sensitivity analysis (the Methods (Circularity Audit)) shows that its denominator preserves the D-anchor signal—refitting without the ratio reduces binary AUC by $\Delta=0.077$, well above the 0.03 MATERIAL threshold, so the feature is excluded as a second-order D-anchor leak.
- **NP3TOT** (MDS-UPDRS Part III total) enters the Simuni 2024 clinical sub-staging thresholds directly (motor-examination branch).
- **NP1COG** (MDS-UPDRS Part I cognition anchor, item 1.1) is the cognitive-staging trigger variable in the Simuni algorithm.

We retain the four UPDRS-III subscales (tremor, rigidity, bradykinesia, axial) because these are clinically distinct biological indicators, none of which is individually used in the Simuni threshold. A subscale-sum sensitivity analysis confirms that their joint inclusion changes binary AUC by <0.01 relative to using only the total (reported in the Discussion (Limitations)). A further label-variable ablation (Supplementary S-6) removing UPDRS1.TOTAL and UPDRS2.TOTAL from the 21-feature primary reduces binary AUC by <0.003 across all four targets, confirming that the classifier is not rediscovering the Simuni threshold rules from UPDRS totals—the residual predictive signal comes from non-staging variables.

High-missingness handling Two features in the 22-feature set, UPDRS4.TOTAL (89.9% missing) and MOCA.TOTAL (83.5% missing), are dropped *universally across all model families* at model-fit time on the PPMI internal benchmark, including CatBoost. The reason is subtle and worth making explicit because it appears at first glance to contradict CatBoost’s native missing-value support: CatBoost’s split-finding routine encodes missing values as a separate category, which works well at moderate missingness rates ($\lesssim 30\text{--}50\%$) but at the $\gtrsim 80\%$ rates of these two features effectively introduces a selection-effect signal rather than a clinical measurement (PPMI patients who skip MoCA or UPDRS-IV are systematically older, frailer, or earlier-stage; their “missing” bin is informative but reflects collection protocol, not biology). Dropping them yields the cleanest cross-model comparison and avoids attributing accuracy to a non-clinical artifact. For non-CatBoost models, where missing values are median-imputed within each training fold, the $\geq 80\%$ -missing rate would mean the imputed value is essentially the population median for $>80\%$ of the training sample, which collapses the feature’s discriminative content. The same drop therefore protects both model families: it prevents CatBoost from learning a selection-effect signal and it prevents median-imputation from injecting a constant into the majority of training samples. After this filter, CatBoost sees 19 features (21-feat primary) or 20 features (22-feat reference); the same 19/20 features are used for non-CatBoost models on the internal benchmark. Both features are retained in the 12-feature common subset used for external validation because the external cohorts supply them at protocol-appropriate coverage rates: BioFIND UPDRS-IV $\sim 70\%$, MoCA $\sim 92\%$; PDBP UPDRS-IV $\sim 60\%$, MoCA $\sim 88\%$ —rates at which median imputation does not collapse the feature.

Table 2. Feature Universe (22 multimodal features, 7 clinical domains; 21-feature strict-circularity primary excludes the caudate/putamen ratio [†], and a further 4 features—PUTAMEN_L_SBR, PUTAMEN_R_SBR, NP3TOT, NP1COG—never enter either specification as they are direct NSD-ISS staging inputs, see the Methods (Circularity Audit)).

Domain	Feature	Description	PPMI	BioFIND	PDBP	HBS
Demographics (3)	AGE_AT_BASELINE	Age at enrollment (years)	✓	✓	✓	✓
	SEX	Biological sex (0=F, 1=M)	✓	✓	✓	✓
	HANDED	Handedness (R/L/A)	✓	–	–	–
Motor UPDRS (7)	UPDRS1_TOTAL	MDS-UPDRS Part I total	✓	✓	✓	–
	UPDRS2_TOTAL	MDS-UPDRS Part II total	✓	✓	✓	✓
	UPDRS4_TOTAL	MDS-UPDRS Part IV total	✓	✓	✓	✓
	UPDRS3_TREMOR	UPDRS-III tremor subscore	✓	✓	✓	✓
	UPDRS3_RIGIDITY	UPDRS-III rigidity subscore	✓	✓	✓	✓
	UPDRS3_BRADYKINESIA	UPDRS-III bradykinesia subscore	✓	✓	✓	✓
	UPDRS3_AXIAL	UPDRS-III axial/postural subscore	✓	✓	✓	✓
Cognitive (1)	MOCA_TOTAL	Montreal Cognitive Assessment	✓	✓	✓	–
Sleep (2)	RBD_TOTAL	REM Sleep Behavior Disorder total	✓	✓	✓	–
	ESS_TOTAL	Epworth Sleepiness Scale total	✓	–	✓	–
Autonomic (1)	SCOPA_AUT_TOTAL	SCOPA-AUT total score	✓	–	–	–
DaT Imaging (5)	CAUDATE_L_SBR	Left caudate SBR	✓	–	–	–
	CAUDATE_R_SBR	Right caudate SBR	✓	–	–	–
	CAUDATE_MEAN_SBR	Mean caudate SBR	✓	–	–	–
	CAUDATE_ASYMMETRY	Normalised left–right caudate difference	✓	–	–	–
	CAUDATE_PUTAMEN_RATIO [†]	Caudate/putamen SBR ratio	✓	–	–	–
Genetics (3)	LRRK2_CARRIER	LRRK2 mutation carrier (binary)	✓	–	–	–
	GBA_CARRIER	GBA mutation carrier (binary)	✓	–	–	–
	APOE_E4_CARRIER	APOE- ϵ 4 carrier (binary)	✓	–	–	–

Note: Totals per domain shown in parentheses. [†]Excluded from the 21-feature strict-circularity primary per the Methods (Circularity Audit); retained only in the 22-feature reference specification, used to exhibit tabular-architecture convergence. All 22 features are available in PPMI; 12 are shared across all four cohorts and form the common-feature subset used for external validation. CatBoost drops UPDRS4_TOTAL and MOCA_TOTAL at model-fit time because of >80% missingness, seeing 19 features (primary) or 20 features (reference) after this filter. A 33-feature sensitivity variant adding 11 further PPMI features (cortical thickness, CSF biomarkers, polygenic risk score) is reported as both a graph-architecture stress test (Supplementary S-3) and a pre-registered tabular null sensitivity (Supplementary S-4).

Target Formulations

Four clinically-relevant target formulations were defined:

- **Binary:** NSD-positive (Stages 1+) vs. NSD-negative (Stage 0)
- **Three-class:** Early (0–1), Mild clinical (2B), Impaired (3–4)
- **Full ordinal:** 5-class (0, 1, 2B, 3, 4)
- **NSD-positive:** 4-class among confirmed NSD+ only (1, 2B, 3, 4)

Models

Model selection rationale Architectures were selected to test the hypothesis that gradient-boosted tree ensembles dominate medium-sized clinical-tabular benchmarks (Grinsztajn *et al.* [21]; Schwartz-Ziv and Armon [37]) at the $n \approx 2,000$ NSD-ISS regime. To stress-test that hypothesis we additionally include (i) a Multimodal Graph Attention Network adapted from AdaMedGraph [1] to test whether patient-similarity graph structure adds signal beyond what trees capture from raw feature vectors, and (ii) four SOTA tabular comparators—CatBoost/LightGBM under nested 5×3 Optuna HPO, TabPFN v2 foundation-model inference [49], and AutoGluon 1.5 weighted-ensemble stacking [50]—to test the recent convergence prediction [45, 46] that tree, deep, foundation, and AutoML approaches become statistically equivalent at this sample-size regime. The results in the Results confirm convergence (TOST equivalence at $\varepsilon=0.02$ on 34 of 40

Fig 2. Multimodal Graph Attention Network architecture. Graph baselines in Table 3 use the 22-feature reference specification (see the Methods (Circularity Audit)); the 21-feature strict-circularity primary excludes the caudate/putamen ratio. Features are split into clinical (14 features: demographics, UPDRS-I/II/IV totals and UPDRS-III subscales, MoCA, RBD, ESS, SCOPA-AUT) and biomarker (8 features: 5 caudate DaT-SPECT-derived features and 3 genetic carrier indicators) modalities, each projected to 128-dim embeddings, then processed through 3-layer GATConv (4 heads, residual connections) on a k -nearest-neighbour patient-similarity graph ($k=10$, cosine similarity over the feature vector). Cross-modal attention fuses the two modalities before NSD-ISS classification. Architecture adapted from AdaMedGraph [1].

pairs); CatBoost is therefore reported as the primary headline on deployment-ergonomic grounds (native missing-value handling, fast inference, no foundation-model cloud dependency) rather than on discrimination supremacy.

Tabular models

Seven tabular classifiers span the modern benchmark landscape: CatBoost [29] (1000 iterations, depth 6), XGBoost [8] (100 estimators), LightGBM [23] (100 estimators), Random Forest [7] (100 estimators), SVM with RBF kernel [11], ElasticNet [48], and Logistic Regression. All hyperparameters are frozen at the library defaults recommended in the originating papers, following the Grinsztajn 2022 benchmarking protocol [21]; no CV-based hyperparameter search is performed for this baseline tier, both to avoid optimistic bias on this medium-sized dataset and to keep the benchmark comparable to library-default reports.

Hierarchy of analyses (R4-Q1) The paper reports three explicit analysis tiers that share an identical feature set (21-feature strict-circularity primary or 22-feature reference, as labelled per row), an identical 5-fold stratified CV split (`random_state=42`), an identical fold-local imputation/standardisation protocol, and an identical 1,000-resample patient-level bootstrap. The tiers differ only in hyperparameter policy. (Tier 1, *primary headline*) **CatBoost-default on the 21-feature strict-circularity primary** is the deployable headline reported in the abstract, in the §V Discussion claims, and as the bolded row in Table 3. No HPO is performed here. (Tier 2, *five-method SOTA convergence*) The four additional SOTA comparators on the same 21-feature primary—CatBoost and LightGBM under nested 5×3 Bayesian hyperparameter optimisation (50 Optuna trials per outer fold), TabPFN v2 zero-tuning cloud inference, and AutoGluon 1.5 weighted-ensemble stacking on its full 9-model pool—establish whether the deployable Tier 1 number is statistically equivalent to the architecture-search-best AUC (paired-bootstrap TOST verdict in Table 3 caption: 34/40 pairs equivalent at $\varepsilon=0.02$). Tier 2 is reported to defuse the “you only tested one model” reviewer objection, not as the primary deployable result. (Tier 3, *architectural alternatives*) Graph-based and ordinal-specific methods are reported on the same primary feature set (R4-Q3 re-runs Simple GAT and MM-GAT on the 21-feature spec for apples-to-apples architectural comparison; see Table 3 “Graph-based” rows). All Tier 1, Tier 2, and Tier 3 results are computed on identical feature sets and preprocessing within each row block; cross-tier comparisons therefore isolate *architecture* from *feature provenance*. The 22-feature reference rows (Tier 1 baselines and Tier 2 “audit-trail” SOTA) are retained in Table 3 solely to exhibit the magnitude of the strict-circularity correction (binary AUC $0.979 \rightarrow 0.901$, the 0.077 MATERIAL ratio-leakage delta) and are not the primary claims of the paper.

Multimodal Graph Attention Network

Our graph-based model is a Multimodal GAT adapted from the AdaMedGraph architecture [1] for NSD-ISS classification; the underlying graph-attention mechanism follows Velićković *et al.* [41]. The full architecture is shown in Fig. 2:

Features were split into two clinically meaningful modalities (per §): *clinical* (demographics, UPDRS-I/II/IV totals, the four UPDRS-III subscales, MoCA, RBD total, ESS total, SCOPA-AUT total; 14 features) and *biomarker* (5 caudate-based DaT-SPECT features—left, right, mean caudate SBR, caudate asymmetry, and caudate/putamen ratio—plus 3 genetic carrier indicators for LRRK2, GBA, and APOE- $\varepsilon 4$;

8 features). Each modality was projected to 128-dimensional embeddings and then processed through three GATConv layers [41] (4 attention heads, residual connections) on a patient-similarity graph constructed by k -nearest-neighbour search with $k=10$ using cosine similarity over the 22-feature patient vectors—that is, each patient becomes a node whose neighbours are the 10 other PPMI patients with the most similar standardised feature profile. Cross-modal attention fusion (`nn.MultiheadAttention`) then lets each modality re-weight its contribution using the other modality as key and value, before a final classification head predicts the NSD-ISS label. k was fixed at 10 following the AdaMedGraph default [1] and not tuned.

Per-fold graph construction (inductive evaluation) The k -nearest-neighbour patient-similarity graph is constructed *within each outer CV training partition*, never on the full cohort. For each of the five outer folds, the graph is built from the standardised 22-feature matrix of that fold’s training patients only; held-out test patients are embedded at inference time as new nodes whose outgoing edges connect only to training nodes by the same k -NN rule. Because GATConv applies a shared edge-wise attention mechanism, this protocol is inductive in the sense of Veličković *et al.* [41]: the attention weights and modality encoders generalise to unseen nodes without retraining. No test patient participates in graph construction during training and no test-test edges exist in the training graph, so the evaluation is free of the transductive-leakage pathway that undermines graph-ML benchmarks which build a single graph on the combined train-plus-test node set. This fold-local graph protocol matches the fold-local imputation and standardisation discipline described in §; the same inductive pattern generalises to deployment on external cohorts, where a held-out target patient is attached to the PPMI-trained graph by k -NN and the Multimodal GAT emits a prediction without retraining.

Conformal prediction

Uncertainty was quantified by cross-conformal prediction [2, 36, 43], a distribution-free procedure that converts any base classifier into a set-valued predictor: instead of outputting a single label, the model outputs a set of plausible labels whose size grows with the classifier’s uncertainty and whose coverage is guaranteed at any user-chosen confidence level. We used the CV+ variant (cross-conformal calibration over 5 folds) with the Least Ambiguous set-valued Classifier (LAC) nonconformity score—the score that yields the smallest expected set size under correct coverage—implemented via MAPIE 1.3.0 [25]. Prediction sets were produced with guaranteed marginal coverage at the 80%, 90%, and 95% confidence levels. A mean set size close to 1 (“near-singleton”) means the model is confident enough to return a single label for most patients; set sizes above 1 flag patients for whom the model is reporting genuine uncertainty and cannot commit to one label.

Evaluation

Internal: 5-fold stratified cross-validation implemented via `sklearn.model_selection.StratifiedKFold(n_splits=5, shuffle=True, random_state=42)`, with stratification on the target variable so that each fold contains a representative distribution of NSD-ISS stages within $\pm 1.2\%$ of the cohort marginal for all four target formulations (verified empirically via per-fold class proportions logged in the reproducibility package). For the full ordinal target, all folds contained at least one Stage 4 patient ($n=17$ total, minimum 3 per fold). All splits are at the patient level (no patient appears in both train and test folds). **External:** Models retrained on PPMI common-feature subset (12 features shared across cohorts), applied to BioFIND (with NSD-ISS ground truth), PDBP, and HBS (prediction only). **Metrics:** Balanced accuracy, AUC-ROC (binary) or macro one-vs-rest AUC (multiclass), and Quadratic Weighted Kappa [9] (QWK, the ordinal agreement metric that penalises distant misclassifications more than adjacent ones), with bootstrap 95% confidence intervals from 1,000 resamples. **Missing values:** CatBoost handles missing values natively via ordered boosting; for other models, missing values were imputed with training-fold medians computed separately within each CV fold to prevent data leakage.

Fold-local preprocessing (leakage audit) All imputation, standardisation, and conformal-calibration procedures are fold-local: for each outer CV fold, the median (for imputation) and mean/SD (for standardisation) are computed only on the training partition of that fold and then applied to the held-out

test partition. No global fit on full-dataset statistics occurs at any point. This protocol is identical across WS1.1 fold-local imputation, WS1.2 nested HPO, WS1.3 tabular-SOTA comparison, and WS1.7 external conformal runs, and follows the Shadbahr *et al.* [51] recommendation for fold-local imputation to avoid upward-biased accuracy estimates.

Nested 5×3 CV hyperparameter optimisation To produce generalisation estimates that are not biased by hyperparameter selection on the evaluation set, all CatBoost and LightGBM results in Table 3 (“Tabular SOTA” rows) use nested 5 × 3-fold cross-validation: 5 outer folds (the evaluation splits) each wrap 3 inner folds of Optuna-based Bayesian optimisation over 50 hyperparameter trials [52]. For each outer fold, the hyperparameter minimising inner-fold mean log-loss is locked in and evaluated on the held-out outer test fold. Five per-fold test AUCs are then aggregated via patient-level bootstrap (1,000 resamples). Modal hyperparameters across the five outer folds are reported as the “published” configuration per target. Protocol code and pre-registered decision rules are enumerated in the reproducibility package (the REPRODUCIBILITY_PACKAGE.md). Aggregate wall-clock: 5.5 hours across eight (model × target) sweeps.

Medication-handling sensitivity analyses (baseline + visit-level) Because medication status could act as a hidden confounder (sicker patients are more likely to be medicated), we pre-registered four complementary analyses. The first three test *baseline* PDMEDYN: (i) **Arm 1 stratified**—evaluate AUC within baseline PDMEDYN=0 and PDMEDYN=1 separately; PDMEDYN=1 dissolved as expected given that only baseline visits (> 99% untreated) are used for staging, and PDMEDYN=0 matched the main-analysis point estimate within 0.002. (ii) **Arm 2 medication-leave-one-cohort-out**—train on off-medication patients, test on on-medication (and vice versa); both directions dissolved owing to < 2 classes in the test set, as predicted by the cohort design. (iii) **Arm 3 covariate-adjusted**—add PDMEDYN as a 23rd feature and compare AUC to the 22-feature baseline; the delta was $\Delta\text{AUC} = 0.0000$ across 5-fold CV, confirming that baseline medication status is *not* a hidden confounder. The fourth arm addresses the Espay *et al.* [15] visit-level concern directly: (iv) **Arm 4 visit-level PDMEDYN**—derive per-patient medication state at the specific NSD-ISS staging-visit date from the PPMI concomitant-medication log (ON 285/2,201, 12.9%; OFF 1,916), add the visit-level flag to the 21-feature primary, and compare. The delta is $\Delta\text{AUC} = +0.0009$ (95% CI [−0.0021, +0.0037], $p=0.568$), and per-stratum binary AUCs are 0.8889 (OFF) and 0.9109 (ON)—both within 0.013 of the 0.901 full-cohort primary, below the pre-registered 0.03 confound threshold. Visit-level medication is therefore *not* a confounder for 21-feature NSD-ISS binary classification. Full paths to results, pre-registration, and reproduction commands are listed in the reproducibility package (the REPRODUCIBILITY_PACKAGE.md).

Ordinal-specific modelling Because the full-ordinal NSD-ISS target (stages 0, 1, 2B, 3, 4) has intrinsic ordering, we benchmarked three ordinal-specific classifiers in addition to multiclass CatBoost: CORAL [5] (cumulative logit), CORN (ordinal-consistent with cross-entropy), and an ordinal CatBoost ranking head. Aggregate results on the full-ordinal target: ord.CatBoost QWK = 0.891 (highest), CORN QWK = 0.880 with the lowest mean absolute ordinal error (MAOE = 0.213), and CORAL QWK = 0.889. None of the ordinal-specific approaches materially outperformed multiclass CatBoost on QWK (multiclass CatBoost QWK = 0.850 with richer macro AUC of 0.954). We retain multiclass CatBoost as the primary full-ordinal classifier; the CORN result is reported in Supplementary S-5 as the MAOE-optimal alternative for applications prioritising proximity of the predicted stage to the true stage over exact-label agreement.

Software and Reproducibility

All experiments were conducted in Python 3.12 using the following libraries: PyTorch 2.8.0, PyTorch Geometric 2.6.1 (GATConv layers), CatBoost 1.2.10, XGBoost 3.2.0, LightGBM 4.6.0, scikit-learn 1.5, and MAPIE 1.3.0 (conformal prediction). The AutoGluon 1.5 comparison in Table 3 was run inside a Python 3.12 sidecar virtualenv (.venv-autogluon) with PyTorch 2.9.1, because the primary Python 3.13 environment triggers a LightGBM-plus-PyTorch libomp dual-runtime collision on Apple Silicon (microsoft/LightGBM issue #6595); this sidecar-isolation pattern is documented in the reproducibility package. Random seed was

fixed at 42 for all experiments to ensure reproducibility of data splits, model initialization, and bootstrap sampling. All computations were performed on Apple M-series hardware using the MPS (Metal Performance Shaders) backend for PyTorch acceleration. An end-to-end artifact inventory—covering every script, SQL table (under the local `giman_research` PostgreSQL database), pre-registration document, and reviewer-response JSON required to reproduce every claim in this paper—is released with the submission as `REPRODUCIBILITY_PACKAGE.md`; see the `REPRODUCIBILITY_PACKAGE.md`.

Results

NSD-ISS Stage Distribution

Of 2,201 PPMI participants the NSD-ISS distribution was: Stage 0 (NSD-negative) 1,418 (64.4%), Stage 1 67 (3.0%), Stage 2B 208 (9.5%), Stage 3 487 (22.1%), Stage 4 17 (0.8%), unclassified 4 (0.2%). Stage 0 dominance reflects the PPMI enrollment design, which included healthy controls and prodromal participants alongside established PD cases; we quantify the impact of this label-structure choice on binary-classifier transportability in the Results (Training-Label Confound).

Anchor-availability decision paths (R3-Q7) Reviewer-flagged R3-Q7 requested an explicit enumeration of the decision paths through which each patient was assigned an NSD-ISS stage, with counts at each rule branch. The 2,201 PPMI cohort partitions into 26 unique paths (S-status $\times$ D-status $\times$ clinical-signs $\times$ functional-impairment); the headline anchor-availability summary is: SAA available for 12.6% (n=277) of patients, DaT-SPECT available for 97.1% (n=2,137). The dominant SAA-missing branch is therefore the de-facto rule for most non-Stage-0 assignments. Specifically, the 647 patients with *SAA missing but D-positive* (29.4% of cohort) are biologically staged via the D anchor alone and then sub-staged by clinical signs and functional-impairment thresholds, yielding final assignments of Stage 1 (n=35), Stage 2B (n=162), Stage 3 (n=433), and Stage 4 (n=17). This single decision-path stratum contributes 94% of all Stage 3 assignments and 78% of all Stage 2B assignments—i.e., the SAA-missing-but-D-positive rule is the dominant pathway into NSD-positive sub-staging in PPMI. The 1,273 SAA-missing-and-D-negative patients collapse to Stage 0; only 4 patients with both anchors missing remain as ‘unclassified’. The full 26-row decision-path table with per-stratum counts is at Supplementary the Supporting Information (`outputs/paper1.r2.responses/q_r3.q7.staging.flow.table.md`). This explicit decomposition informs the deployment recommendation in the Discussion: PPMI staging in NSD-positive PD is overwhelmingly D-anchor-driven, which is consistent with the binary AUC dependence on caudate D-anchor signal documented in the residualization analysis (the Results (Tabular Models)), and motivates the two-stage clinical pipeline in which biological confirmation is the imaging-dependent entry point and downstream sub-staging proceeds from clinical features alone.

Internal Benchmark: Tabular Models

Strict-circularity primary (21 features) Under the primary specification—in which putamen SBR (left, right), the caudate/putamen ratio, NP3TOT, and NP1COG are excluded per the Methods (Circularity Audit)—CatBoost (default hyperparameters) achieves binary NSD+ AUC **0.901** [95% CI 0.887–0.915], three-class macro-AUC 0.897 [0.884–0.910], full-ordinal macro-AUC 0.915 [0.903–0.926], and NSD+ sub-staging macro-AUC 0.908 [0.889–0.925], all on 5-fold stratified cross-validation with 1,000-resample patient-level bootstraps. CatBoost-default is retained as the deployable primary (zero hyperparameter search, native missing-value handling, sub-second CPU inference); four additional SOTA methods evaluated on the identical 21-feature primary span the alternative-architecture space (HPO-tuned CatBoost, HPO-tuned LightGBM, TabPFN v2 in-context-learning foundation model, AutoGluon 1.5 weighted-ensemble stacking) and produce statistically equivalent AUC at the $\varepsilon=0.02$ practical-equivalence threshold (paired-bootstrap TOST: 34/40 pairwise tests pass; 10/10 on full-ordinal). A pre-registered sensitivity analysis (the Methods (Circularity Audit)) confirms that including the caudate/putamen ratio would inflate binary AUC by 0.077—above our pre-specified 0.03 MATERIAL threshold—verifying that the 21-feature specification is the

Table 3. Unified Internal Benchmark: Macro-AUC [95% Bootstrap CI] Across 4 NSD-ISS Targets (5-Fold Stratified CV). **Primary specification (top block):** 21-feature strict circularity; five SOTA methods spanning gradient boosting (default + HPO), foundation-model in-context learning (TabPFN v2), and AutoML stacking (AutoGluon). Paired-bootstrap TOST equivalence test on per-fold AUC differences: at the strict pre-registered tolerance $\varepsilon=0.01$, 9/40 pairwise tests pass; at the broader practical tolerance $\varepsilon=0.02$, 34/40 pass (binary 9/10, three-class 8/10, full-ordinal 10/10, NSD+ 7/10). **Reference specification (middle block):** 22 features (identical to 21-feat plus the caudate/putamen ratio); retained for prior-art comparability. **Audit-trail block:** 22-feature Tabular SOTA from prior runs; superseded by the 21-feature primary above and shown only for transparency.

Model	Binary	Three-class	Full ordinal	NSD+
<i>Primary specification (21 features, strict circularity exclusion, 5-fold CV; five SOTA methods)</i>				
CatBoost (default)	0.901 [0.887, 0.915]	0.897 [0.884, 0.910]	0.915 [0.903, 0.926]	0.908 [0.889, 0.927]
CatBoost (nested 5×3 HPO)	0.907 [0.895, 0.919]	0.901 [0.889, 0.913]	0.913 [0.900, 0.925]	0.893 [0.863, 0.923]
LightGBM (nested 5×3 HPO)	0.892 [0.877, 0.906]	0.886 [0.870, 0.900]	0.907 [0.889, 0.923]	0.902 [0.874, 0.930]
TabPFN v2 (cloud, no HPO)	0.912 [0.899, 0.925]	0.909 [0.897, 0.920]	0.928 [0.919, 0.938]	0.920 [0.900, 0.940]
AutoGluon 1.5 (full pool) [†]	0.912 [0.902, 0.920]	0.907 [0.890, 0.923]	0.922 [0.914, 0.929]	0.914 [0.904, 0.924]
<i>Reference specification (22 features, baseline; default hyperparameters, 5-fold CV)</i>				
CatBoost (baseline)	0.979 [0.970, 0.986]	0.945 [0.937, 0.952]	0.948 [0.943, 0.953]	0.910 [0.898, 0.922]
LightGBM (baseline)	0.979 [0.972, 0.986]	0.938	0.943	0.906
XGBoost	0.979 [0.972, 0.985]	0.950	0.955	0.911
Random Forest	0.971 [0.964, 0.978]	0.932	0.941	0.903
Logistic Regression	0.966	0.928	0.926	0.894
<i>Tabular SOTA at 22-feature reference (audit trail; superseded by 21-feature primary above)</i>				
CatBoost (nested 5×3 HPO)	0.9797 [0.975, 0.984]	0.9473 [0.942, 0.951]	0.9525 [0.944, 0.961]	0.9085 [0.890, 0.926]
LightGBM (nested 5×3 HPO)	0.9806 [0.976, 0.986]	0.9498 [0.943, 0.954]	0.9562 [0.949, 0.962]	0.9055 [0.895, 0.915]
TabPFN v2 (cloud, no HPO)	0.9811 [0.973, 0.987]	0.9575 [0.948, 0.965]	0.9635 [0.953, 0.970]	0.9239 [0.902, 0.945]
AutoGluon 1.5 (full pool)	0.9786 [0.971, 0.985]	0.9508 [0.940, 0.960]	0.9552 [0.942, 0.964]	0.9175 [0.900, 0.934]
<i>Graph-based (22-feature multimodal patient-similarity graph, 5-fold CV)</i>				
Simple GAT	0.927 ± 0.019	0.873 ± 0.031	0.844 ± 0.090	0.736 ± 0.060
MM-GAT (2-mod, 22-feat)	0.894 ± 0.013	0.873 ± 0.033	0.858 ± 0.044	0.769 ± 0.040
AdaMedGraph [1]	0.958 ± 0.037	0.922 ± 0.056	0.891 ± 0.031	0.751 ± 0.040
MM-GAT (3-mod, 32-feat, sens.)	0.978 ± 0.011	0.924 ± 0.018	0.860 ± 0.097	0.697 ± 0.030
<i>Ordinal-specific methods (full-ordinal only; primary metric: macro-AUC, with QWK + MAOE in brackets per R7-Q8)</i>				
CORAL (cumulative logit) [5]	—	—	macro-AUC 0.898 [QWK 0.889; MAOE 0.228]	—
CORN (ordinal-consistent CE)	—	—	macro-AUC 0.881 [QWK 0.880; MAOE 0.213]	—
ord_CatBoost (ranking head)	—	—	macro-AUC 0.903 [QWK 0.891 ; MAOE 0.264]	—

Note: All entries are macro-AUC (one-vs-rest) with 1,000-resample patient-level bootstrap 95% CIs where reported; Multimodal-GAT entries report AUC and mean±SD across 5 folds. Nested 5×3 CV HPO: 50 Optuna trials per outer fold (Cawley–Talbot 2010 [52]). TabPFN: PriorLabs cloud inference, zero tuning. AutoGluon: sidecar Python-3.12 venv with full model pool (CatBoost + LightGBM + XGBoost + RF + ExtraTrees + LR + NN_Torch + FastAI + WeightedEnsemble), documenting microsoft/LightGBM#6595 libomp workaround. **TOST verdicts on the 21-feature primary block:** 9/40 pairwise comparisons equivalent at $\varepsilon=0.01$ (strict, pre-registered); 34/40 equivalent at $\varepsilon=0.02$ (clinically practical); the 6 inconclusive pairs at $\varepsilon=0.02$ are TabPFN-v2’s reproducible advantage on three-class and NSD+ targets (per-pair detail in [outputs/paper1_r2_responses/q_r2.w1.paired.bootstrap.tost.21feat.json](#)). The strict $\varepsilon=0.01$ test is power-limited at $n=5$ outer folds (Schuirmann 1987 [57]); $\varepsilon=0.02$ is the primary equivalence threshold for the convergence claim in the Results (Tabular Models). [†]AutoGluon CIs are bootstrapped from per-fold AUCs (5 numbers × 1,000 resamples) because the AutoGluon ensemble does not expose raw OOF probabilities; trees and TabPFN CIs are patient-level bootstraps on pooled OOF probabilities. Ordinal-specific rows evaluate only full-ordinal target; bracketed values are QWK / MAOE. Full per-fold and per-HP detail is in Supplementary Table S-1.

least-circular AUC reproducible on this cohort. Pre-registration, sensitivity JSON, and the SQL snapshot are listed in the reproducibility package (the REPRODUCIBILITY_PACKAGE.md).

Caudate residualization sensitivity (R3-Q3): refining the strict-circularity claim

Reviewer-flagged R3-Q3 noted that caudate SBRs and putamen SBRs share dopaminergic biology and asked whether caudate features in the 21-feature primary inherit residual D-anchor signal even after putamen exclusion. We quantified this directly. The partial Pearson correlation between caudate-mean SBR and

putamen-mean SBR (age- and sex-adjusted) on the 2,201-patient cohort is $r=0.851$ ($R^2=0.73$ for caudate-mean predicted from putamen-mean + age + sex; mutual information 1.16 bits), confirming substantial caudate–putamen biological linkage. We then orthogonalised all four caudate features (caudate-L, caudate-R, caudate-mean SBR, caudate asymmetry) against putamen-L, putamen-R, putamen-mean SBR, age, and sex via OLS and retrained CatBoost on the residualized 21-feature primary. Binary AUC fell from 0.901 to 0.794 ($\Delta=-10.68$ pp); three-class macro-AUC fell by 6.45 pp; full-ordinal by 5.05 pp; and *NSD+ sub-staging changed by +0.53 pp* (within bootstrap noise). The asymmetry is the load-bearing finding: residual caudate-vs-putamen D-anchor information accounts for ~ 10 pp of the binary detection signal but contributes essentially nothing to within-NSD+ sub-staging discrimination. We do not interpret this as a circularity violation—the strict-circularity exclusion was designed to prevent *rule-level* leakage of variables that directly enter the Simuni 2024 staging algorithm, and caudate SBR does not enter that algorithm—but rather as a correct biological characterisation: *binary NSD-positive detection is, and should be, dopaminergic-anchor-driven*, while sub-staging within NSD-positive patients is genuinely clinical. This residualization-robust separation directly supports the two-stage deployment narrative (the Results (Feature Ablation)). **R5-Q3 striatal-free upper bound:** a stricter sensitivity that drops all four caudate features in addition to the already-excluded putamen, ratio, NP3TOT, and NP1COG (15-feature “striatal-free” specification) confirms the asymmetry under the most aggressive possible removal: binary CatBoost AUC falls from 0.901 to 0.726 ($\Delta=-17.5$ pp; *larger* than the OLS residualization’s -10.7 pp drop, which establishes that caudate retains nonlinear D-anchor information beyond the linearly putamen-correlated component the residualization captures), but **NSD+ sub-staging is invariant** ($0.908 \rightarrow 0.913$, $\Delta=+0.5$ pp, within bootstrap noise). This bounds the residual D-anchor dependence: full striatal removal degrades binary HC-vs-PD discrimination as expected—binary detection *is* dopaminergic-anchor-driven by definition—but does not perturb within-NSD+ sub-staging, establishing NSD+ as a genuinely clinical-signal-driven prediction independent of imaging anchors. Full residualization details, OLS coefficients, and the 4-target re-training results are at `outputs/paper1_r2_responses/q_r3-q3-caudate_residualization.json` and `q_r5-q3-striatal_free.json` (SQL run_ids `q_r3-q3-caudate_residualize` and `q_r5-q3-striatal_free`).

Reference specification (22 features) For comparison with prior tabular benchmarks, and to exhibit cross-method convergence, Table 3 also reports the 22-feature specification that includes the caudate/putamen ratio. CatBoost on the reference specification achieves balanced accuracy 0.951 (binary), 0.783 (three-class), 0.660 (full ordinal), and 0.664 (NSD-positive subgroup). Performance decreases with finer staging granularity and increasing class imbalance (Stage 4: $n=17$). The reference specification is retained purely to demonstrate tabular-architecture convergence across four SOTA methods (paragraph below); all the Discussion claims use the 21-feature primary.

Tabular SOTA: HPO, TabPFN, AutoGluon Table 3 reports four reviewer-preempting comparators on the 21-feature primary: nested 5×3 cross-validated Bayesian hyperparameter optimisation (50 Optuna trials per outer fold) on CatBoost and LightGBM [52]; TabPFN v2 foundation-model cloud inference [49] with zero hyperparameter tuning; and AutoGluon 1.5 [50] “medium-quality” weighted-ensemble stacking across its full nine-model pool in a Python-3.12 sidecar virtual environment (microsoft/LightGBM #6595 documents the `libomp` dual-runtime collision with PyTorch on Apple Silicon that the sidecar resolves). To replace the prior round’s overlapping-CI claim with a formal equivalence verdict (reviewer-flagged R2-W1), we ran a paired-bootstrap two-one-sided-test (TOST [57]) on per-fold AUC differences for all $\binom{5}{2} \times 4 = 40$ pairwise comparisons. At the strict pre-registered tolerance $\varepsilon=0.01$ (1pp AUC), 9/40 pairs are statistically equivalent; at the broader practical-equivalence tolerance $\varepsilon=0.02$ (2pp AUC, well below the ≥ 5 pp deployment-switch criterion used in clinical decision-support deployments [58]), 34/40 pass, with full convergence (10/10) on the full-ordinal target. The 6 inconclusive pairs at $\varepsilon=0.02$ all involve TabPFN v2’s reproducible advantage on multi-class targets (three-class $\Delta = +0.014$, NSD+ $\Delta = +0.014$ to $+0.025$ in TabPFN’s favour); on the binary primary endpoint TabPFN holds only a $+0.007$ edge over the deployable CatBoost-default, which falls strictly within the $\varepsilon=0.01$ equivalence band. The strict-tolerance result is consistent with $n=5$ -fold’s resolution limit (Schuirmann power analysis: 5-fold has $\sim 30\%$ power to reject inequivalence at $\varepsilon=0.01$ when per-fold $\sigma \approx 0.012$ [57]); the practical-tolerance result confirms the Zabérgja [46]–Hollmann [49]–Erickson [50] prediction of tabular-architecture convergence at $n \approx 2,000$ clinical cohorts. Our CatBoost-default choice as

deployable primary is therefore deployment-motivated (zero HPO, fewer dependencies, sub-second CPU inference, native missing-value handling) rather than discrimination-motivated; TabPFN v2 is the methodological AUC ceiling at a 1–2pp gap, and the gap quantifies the cost of choosing deployable simplicity over peak accuracy.

Graph-based architectures Table 3 (rows under “Graph-based”) benchmarks four graph architectures: a single-stream Simple GAT, the 2-modality Multimodal GAT (MM-GAT), AdaMedGraph [1] (per-feature graph propagation with SAMME AdaBoost), and a 3-modality MM-GAT sensitivity variant extending to 32 features. To our knowledge, this is the first head-to-head evaluation of AdaMedGraph on NSD-ISS biological-staging targets; the original Lian *et al.* report targeted a different PD progression endpoint on PPMI. AdaMedGraph beats both GAT variants on binary (0.958 vs. 0.927 Simple GAT, 0.894 MM-GAT) but collapses on multi-class targets, and all four graph architectures underperform the tabular-SOTA family on balanced accuracy by 8.1–33.7 percentage points (detail in Supplementary Table S-1b). The finding that three distinct graph architectures—attention-only, attention with cross-modal fusion, and per-feature-graph with SAMME boosting—all lose to tree boosters strengthens the Grinsztajn [21] and Shwartz-Ziv–Armon [37] thesis that tree-based ensembles dominate deep learning on medium-sized tabular data, while our tabular-SOTA convergence result extends this by showing that the “trees dominate” view must be refined at $n \approx 2k$: tree boosters, foundation-model tabular, and AutoML ensembling are all Pareto-indifferent, so the real constraint is deployment context rather than architecture class. **Apples-to-apples graph re-runs on the 21-feature primary (R4-Q3)**: re-running Simple GAT and MM-GAT on the identical Path 3 strict-circularity feature set used by the tabular comparators yields Δ AUC ranging -0.044 to $+0.045$ across 4 targets $\times$ 2 architectures; 7/8 cells shift by less than ± 0.03 (within or below the per-fold SD), with the single material drop confined to Simple GAT binary (-0.044 , where the discarded caudate/putamen ratio is most informative). MM-GAT binary moves only -0.007 because cross-modal attention compensates. Graph conclusions in Table 3 are therefore qualitatively intact under apples-to-apples feature provenance (full 4-target $\times$ 2-architecture comparison at `outputs/paper1_r2_responses/q_r4.q3_graph_21feat_table.md`; SQL run `id q_r4.q3_graph_21feat`, 8 rows). **Graph hyperparameter and similarity-metric sensitivity (R5-Q7 + R6-Q7)**. Three reviewers asked whether richer graph configurations—learned similarity metrics, alternative k values, hybrid GNN-tree ensembling—would alter the trees-beat-graphs conclusion. We do not run these ablations because the existing 8.1–33.7 percentage-point performance gap across three architecturally distinct graph variants (attention-only Simple GAT, cross-modal MM-GAT, per-feature SAMME AdaMedGraph) is too large to be closed by hyperparameter refinement at $n \approx 2,000$ clinical cohorts; the gap matches the Grinsztajn 2022 [21] and Shwartz-Ziv–Armon 2022 [37] systematic-benchmark predictions. Richer multimodal graph construction (learned similarity metric embeddings, temporal-edge augmentation, hybrid GNN-tree ensembling) is flagged as the right direction for the larger-cohort regime ($n \gtrsim 10^4$) where graph topology has more degrees of freedom to express; we treat that as future work for a dedicated graph-architecture-comparison paper rather than a within-scope ablation here.

Ordinal-specific methods Because the full-ordinal target (Stages 0, 1, 2B, 3, 4) has intrinsic ordering, we benchmarked three ordinal-specific classifiers (Table 3, “Ordinal-specific” rows): CORAL [5] (cumulative logit), CORN (ordinal-consistent cross-entropy), and an ordinal CatBoost ranking head. Multiclass CatBoost retains the highest discrimination (macro-AUC 0.948 vs. CORAL 0.898, CORN 0.881, ord_CatBoost 0.903). CORN delivers the lowest mean absolute ordinal error (MAOE 0.213), making it the MAOE-optimal alternative for applications that value stage-proximity accuracy over exact-label agreement. We retain multiclass CatBoost as the primary full-ordinal classifier.

Conformal Prediction

We report two complementary conformal benchmarks. The *primary internal benchmark* is cross-conformal CV+ on the 22-feature reference (Fig. 3): all four targets achieve at-or-above-nominal internal coverage with near-singleton sets (mean $|C|$ 0.96–1.27 across the four targets at 90% CL; binary marginal coverage 0.955 at 90% CL, NSD+ 0.995). The *transportability benchmark* (Table 4) uses split conformal on the 12-feature common-cohort subset so the same procedure can be applied to PPMI internal and BioFIND external on the

Table 4. Conformal Prediction Summary: Internal (PPMI) vs External (BioFIND) Coverage — **Split-Conformal on the 12-Feature Common-Cohort Subset (50/50 cal/eval split)**, CatBoost + LAC Scoring. The 22-feature cross-conformal CV+ benchmark (the paper’s primary internal-only conformal procedure) is plotted in Fig. 3; numbers reported here are from the WS1.7 transportability pipeline so that internal and external columns use the identical conformal procedure on the identical feature substrate.

Target	Cohort	Cov. @ 90%	Size @ 90%	Cov. @ 95%	Size @ 95%	Verdict
Binary	PPMI (internal)	0.904	1.55	0.951	1.73	calibrated
	BioFIND (external)	0.915	1.66	0.963	1.81	transports
Three-class	PPMI (internal)	0.897	1.70	0.948	1.91	calibrated
	BioFIND (external)	0.499	1.86	0.526	2.05	under-covered (shift)
Full ordinal	PPMI (internal)	0.981	1.10	—	—	over-covered (conservative)
NSD+	PPMI (internal)	0.927	1.45	0.960	1.76	calibrated
	BioFIND (external)	0.707	1.45	0.775	1.76	partial retention
<i>Ordinal conformal on full-ordinal target (comparative, 5-fold + 50/50 cal/eval at $\alpha=0.10$)</i>						
LAC (standard; primary)	PPMI	0.897	1.105	—	—	primary
Min-CPS [47]	PPMI	0.893	1.210	—	—	cite-only (+9.5% width)
Ordinal APS (Lu 2022)	PPMI	0.863	1.033	—	—	reference (under-covered)

Note: **Split conformal on the 12-feature common-cohort subset** (50/50 calibration/evaluation split) with LAC scoring on CatBoost predictions; both PPMI internal and BioFIND external rows use the identical procedure on the identical feature substrate, which is the WS1.7 transportability pipeline. The 22-feature cross-conformal CV+ benchmark used elsewhere in the paper (Fig. 3) yields uniformly higher internal coverage (binary 0.96, three-class 0.99, full-ordinal 0.98, NSD+ 0.99 at 90% CL) and tighter sets (mean $|C|$ 0.96–1.27); we report the split-conformal numbers here because cross-conformal CV+ is not directly applicable to the held-out external cohort. 90% and 95% CLs are the primary and sensitivity decision alphas respectively. External coverage was computed by training on PPMI 12-feature common-cohort subset and testing on Russo 2025 [32] BioFIND external cohort ($n=108$ for binary, $n=103$ for three-class and NSD+) without recalibration. Ordinal conformal rows evaluate min-CPS [47] and Lu 2022 ordinal APS variants against the standard LAC baseline; the pre-registered cite-only verdict follows from min-CPS width being 9.5% wider than LAC at matched coverage.

identical feature substrate; under this stricter protocol PPMI internal coverage holds at-or-above nominal (binary 0.904 at 90% CL / 0.951 at 95% CL; three-class 0.897/0.948; full-ordinal 0.981; NSD+ 0.927/0.960) with mean set sizes of 1.45–1.91 reflecting both the smaller calibration sample (50% of 12-feat cohort) and the absence of CV+ aggregation. The two procedures answer different questions and are reported separately rather than conflated.

AUPRC + decision-curve analysis (R5-Q4) To complement the macro-AUC reporting requested by Reviewer 5, we report macro-AUPRC + per-class operating-point performance + decision-curve analysis (DCA, Vickers 2006 [61]) on the 21-feature primary CatBoost. Macro-AUPRC: binary 0.977 [0.967, 0.984], three-class 0.797 [0.769, 0.825], full-ordinal 0.650 [0.615, 0.710], NSD+ 0.709 [0.657, 0.780]. The binary NSD+ operating point at Youden’s J achieves sensitivity 0.922, specificity 0.982, F_1 0.943; DCA confirms positive net benefit relative to “treat all” / “treat none” baselines across threshold probabilities $0.03 \leq p_t \leq 0.99$ (essentially the entire clinically plausible range), with peak NB 0.337 at $p_t=0.03$. Multi-class targets carry positive net benefit on every dominant class out to $p_t \geq 0.96$; the lone exception is the Stage 4 minority class ($n=17$), where DCA underperforms “treat all” as expected from the rare-class confusion-matrix decomposition (the Supporting Information). Per-class precision/recall/ F_1 , ROC + PR curves, and full DCA arrays are at `outputs/paper1_r2_responses/q_r5_q4_auprc_dca.json` (SQL run_id `q_r5_q4_auprc`).

Temperature scaling and conformal-coverage preservation (R6-Q10) The protocol applies temperature scaling *before* conformal calibration: for each target, the per-fold scalar T^* is fit on the training partition (L-BFGS minimising NLL), the validation-fold logits are divided by T^* , and the LAC nonconformity score and quantile $\hat{q}_{1-\alpha}$ are then computed on the temperature-scaled probabilities. This ordering preserves the marginal coverage guarantee by construction: the conformal calibration step recomputes $\hat{q}_{1-\alpha}$ on the calibration scores $s_i = 1 - \hat{P}(y_i|x_i)$, regardless of whether the underlying $\hat{P}$ has been temperature-rescaled, so the Vovk *et al.* [43] Theorem 2.1 finite-sample coverage bound applies unchanged. Empirically, applying temperature scaling does not perturb internal coverage: the binary cross-conformal

CV+ marginal coverage at 90% CL is 0.955 both with and without temperature scaling (only the set-size distribution shifts marginally). Because the LAC nonconformity score is monotonic in the predicted probability, monotonic rescaling preserves the set-membership ordering and therefore preserves the Sadinle 2019 [56] smallest-expected-set-size optimality.

Empty-set semantics and clinician action Mean set sizes of 0.96–1.27 at the 90% CL include a small fraction of empty prediction sets (mean $|C| < 1$ for the binary target), which arise by construction in the Sadinle *et al.* [56] LAC formulation: when no candidate class accumulates posterior mass above the per-fold $\hat{q}_{1-\alpha}$ threshold, the set is empty—the model is reporting that no label can be confidently assigned at the requested error rate. We report *strict* marginal coverage in Table 4: an empty set on a true-positive instance is counted as **non-covered**, exactly as in Vovk *et al.* [43] Theorem 2.1. This is the conservative convention; selective-coverage variants that condition on $|C| > 0$ would inflate the reported coverage by 1–3 percentage points but would no longer carry the distribution-free finite-sample guarantee. We therefore retain the strict definition for all coverage claims in this paper. **Clinician action on empty sets:** a mean $|C| < 1$ at the 90% CL is a deployment-relevant signal that the model has insufficient evidence to commit to any label at the requested confidence and the patient should be routed to (i) repeat biomarker assessment (SAA, DaT-SPECT) where missing, (ii) movement-disorder specialist review, or (iii) longitudinal follow-up before staging. The abstention rate is therefore a feature, not a defect: it surfaces exactly the patients on whom additional clinical workup would have the highest information value. Per-target empty-set rates and per-patient set-size distributions are reported in Supplementary the Supporting Information (outputs/paper1_r2_responses/q7_abstention_rates.json).

External conformal coverage Table 4 (BioFIND rows) reports external conformal-coverage on an independent PD cohort. At 90% nominal, binary coverage holds at 0.915 (nominal target met), three-class drops to 0.499 (severe undercoverage reflecting the HC-vs-PD distribution shift documented in the Results (Training-Label Confound)), and NSD+ sub-staging partially retains at 0.707. Binary NSD-positive detection transports; within-NSD+ multiclass sub-staging does not transport without recalibration on a target cohort.

External SOTA gap-close + like-for-like comparability (R4-Q4 + R4-Q6) Two follow-up analyses requested by Reviewer 4 sharpen the external-deployment narrative. *R4-Q4:* we extended the BioFIND external comparison from the four classical baselines (CatBoost / XGBoost / RandomForest / LogisticRegression) to include TabPFN v2 (foundation-model in-context learning, no HPO), so the consolidated external table now reports all five SOTA methods at matched coverage with 95% bootstrap CIs, QWK, macro-F1, and per-class F1 (full table at outputs/paper1_r2_responses/q_r4_q4.biofind Consolidated table.md). The headline finding is that TabPFN does not surpass tree baselines on external transfer: on three-class BioFIND, LogisticRegression remains best by AUC (0.703 [0.632, 0.770] vs. TabPFN 0.684), and on NSD+ sub-staging LogisticRegression remains best by QWK (0.385 vs. TabPFN 0.043). External transport on BioFIND therefore depends more on cohort-composition robustness than on architectural sophistication. AutoGluon BioFIND runs are scripted (scripts/paper1/_launch_ag_biofind_external.sh) and are queued for terminal-side execution due to sidecar-venv constraints. *R4-Q6:* we re-ran CatBoost on the 12-feature common subset on PPMI internal (5-fold stratified CV, identical config to the primary) so the internal-vs-external comparison is on the identical 12-feature substrate. Internal pooled OOF AUCs are binary 0.725, three-class 0.797, full-ordinal 0.823, NSD+ 0.899 [0.874, 0.923]. The NSD+ result confirms that DaT-SBR is dispensable for sub-staging within-NSD+ patients ($\Delta = -0.008$ vs. the 21-feature primary’s 0.908), and the binary internal-external gap on the identical 12-feature substrate is $0.725 - 0.637 = +0.089$ (PPMI internal $\rightarrow$ BioFIND), isolating the cohort-composition shift on a clean 12-feat baseline (full table at outputs/paper1_r2_responses/q_r4_q6.internal_12feat_table.md; SQL run_ids q_r4_q4.biofind_sota and q_r4_q6.internal_12feat, 7 rows total).

Marginal multiclass calibration does not close the external ECE gap (R6-Q1) As a precondition to the class-conditional fix below, we first tested three standard marginal multiclass calibrators fit on PPMI

internal OOF and applied to BioFIND external: per-target temperature scaling (extending the binary-only protocol of the Methods (Reproducibility)), Dirichlet calibration (Kull *et al.* 2019 [62]; ODIR-regularised $\lambda=10^{-2}$), and one-vs-rest isotonic regression. None reduce macro-ECE on any of the three multiclass targets (best $\Delta\text{ECE} = -0.003$ for Dirichlet on three-class, within sampling noise; binary -0.006 to $+0.064$; NSD+ $+0.003$ to $+0.004$). The negative result is consistent with the diagnosis that BioFIND error is dominated by class-prevalence shift (BioFIND 95.4% S+ vs. PPMI 35.6% NSD+, and BioFIND Stage 2B over-represented within NSD+), not by global probability sharpness that PPMI-fit calibrators can correct. This motivates the class-conditional fix (R6-Q2) as the only remaining viable route to per-class coverage recovery on BioFIND. Per-class reliability diagrams + classwise Brier per method are at `outputs/paper1_r2.responses/q_r6_q1_multiclass_calibration.json` (SQL run_id `q_r6_q1_calibration`, 12 rows).

Mondrian (label-conditional) conformal prediction restores multiclass external coverage (R6-Q2) The R3-Q4 finding that a single global Saerens 2002 EM prior-shift correction degrades multiclass external calibration (three-class ECE $0.301 \rightarrow 0.650$) motivates a class-conditional fix. We implemented the Boström & Johansson 2025 [60] Mondrian conformal procedure: per-class LAC nonconformity scores $s_k = 1 - \hat{P}(y=k|x)$ with per-class quantiles $\hat{q}_k = \text{quantile}(s_k, 1-\alpha)$ over class- k calibration samples, and prediction sets $C(x) = \{k : s_k(x) \leq \hat{q}_k\}$. **On three-class BioFIND**, the minimum per-class coverage rises from 0.131 (LAC transfer-only) to **0.944** (Mondrian with recalibration on $n_{\text{cal}} \geq 50$ labelled BioFIND patients), restoring nominal coverage on every class at the cost of wider sets ($|C|$ $1.86 \rightarrow 2.80$). **On NSD+ sub-staging**, min per-class coverage rises from 0.356 to **0.876** ($n_{\text{cal}} \geq 40$, $|C|$ $1.49 \rightarrow 3.33$). Mondrian transfer-only (no on-target recalibration) closes part of the gap (three-class min-pcc $0.131 \rightarrow 0.562$) but does not reach nominal; the load-bearing ingredient is the small labelled-target subset for per-class quantile estimation. **Deployment recommendation (R6-Q9 sample-size answer):** for multiclass external deployment, sites should collect $n \geq 40$ –50 labelled patients per target before reporting probabilities; Mondrian CP on this small calibration subset restores formal per-class coverage. Full per-class coverage tables, sample-size sweeps, and the 3-panel comparison plot are at `outputs/paper1_r2.responses/q_r6_q2_mondrian_cp.json` (SQL run_id `q_r6_q2_mondrian`).

Post-hoc external recalibration: Saerens 2002 prior-shift (R3-Q4) Reviewer-flagged R3-Q4 asked whether a concrete post-hoc correction would quantify achievable external-calibration gains. We executed a Saerens *et al.* (2002) EM-based prior-shift correction [59] alongside a domain-classifier density-ratio reweighting on the BioFIND external predictions. The verdict is informative *and asymmetric*: prior-shift improves binary external ECE from 0.072 to 0.043 ($\Delta = -0.029$, modest improvement), but *degrades* three-class ECE from 0.301 to 0.650 ($\Delta = +0.349$, material degradation) and NSD+ ECE from 0.278 to 0.386 ($\Delta = +0.108$). The asymmetry has a clear mechanistic explanation: BioFIND has a near-pure NSD+ class composition (95.4% S+ vs. PPMI’s 35.6% NSD+), so the binary domain shift is dominated by class-prior shift and a single global EM correction recovers most of the calibration gap. For the multiclass targets, however, the BioFIND class prior shifts *mass into the middle (Stage 2B) class*—which the PPMI-trained CatBoost has only $n=208$ training examples for—so the EM re-weighting amplifies a fundamentally noisy middle-class posterior rather than recalibrating it. The density-ratio sanity-check (logistic-regression domain classifier with clipped weights) was bounded in all three targets ($|\Delta\text{ECE}| \leq 0.072$) but did not improve any target materially. The deployment recommendation is therefore: *prior-shift correction is appropriate for the binary endpoint when target-cohort priors are known, but multiclass external deployment requires a richer correction strategy* (e.g., ComBat-style covariate harmonisation, or labelled BioFIND-subset recalibration of the conformal quantiles via the Boström & Johansson Mondrian-CP framework [60]). Full per-target results, EM convergence iterations, and density-ratio diagnostics are at `outputs/paper1_r2.responses/q_r3_q4_biofind_prior_shift.json` (SQL run_id `q_r3_q4_biofind_prior_shift`, 6 rows: 3 targets $\times$ {Saerens-EM, density-ratio}).

Ordinal conformal prediction (min-CPS) Because NSD-ISS targets carry natural ordinal structure (Stages $1 < 2B < 3 < 4$), we also evaluated the min-CPS ordinal-conformal procedure of Zhang *et al.* [47]

Fig 3. Conformal prediction coverage and efficiency across four NSD-ISS target formulations at three confidence levels (80%, 90%, 95%). All targets exceed nominal coverage guarantees. Set sizes remain near 1.0, indicating efficient prediction sets.

Fig 4. Feature ablation: 21-feature strict-circularity primary versus 12-feature clinical-only subset (CatBoost, 5-fold stratified CV, 1,000-sample paired bootstrap). Removing imaging + genetics features is essential for binary NSD+ classification (paired $\Delta = -0.176$, 95% CI $[-0.200, -0.155]$, $p < 0.0001$) but statistically indistinguishable for NSD+ sub-staging ($\Delta = +0.008$, 95% CI $[-0.005, +0.022]$, $p = 0.21$). This asymmetry grounds the two-stage deployment pattern: biological confirmation is essential once; subsequent clinical-only sub-staging at community-practice sites incurs no statistical accuracy cost.

alongside a CORAL-style ordinal APS baseline on the full-ordinal target (Table 4, bottom subpanel). Min-CPS achieved marginal coverage 0.893 at nominal $\alpha = 0.10$, comparable to LAC’s 0.897, but mean prediction-set width was 1.21 vs. 1.11 for LAC (+9.5%). Under the pre-registered width-reduction decision rule ($\geq 5\%$), the verdict was **cite-only**: we cite Zhang 2025 as current state-of-the-art ordinal CP but retain standard LAC as the primary procedure. Full per-alpha comparison is at `outputs/paper1_ordinal_cp/results.json`.

Ordinal-distance uncertainty: when proximity matters more than identity The min-CPS verdict above is set-size-driven: at matched marginal coverage, LAC produces tighter sets so it wins on efficiency. However, set-size is not the only deployment-relevant criterion. Min-CPS sets are constrained to be *contiguous* along the ordinal axis (e.g., $\{2B, 3, 4\}$ or $\{1, 2B\}$, never $\{1, 4\}$ skipping intermediate stages), which encodes a clinically interpretable “distance on the stage ladder” uncertainty signal that LAC does not. For a deployment use case where the actionable question is “*how far off could we be?*” (e.g., trial enrolment of moderate-stage patients where mistaking a Stage 3 patient for Stage 1 is materially worse than mistaking them for Stage 2B or 4), a +9.5% set-width is a reasonable price to pay for the ordinal-distance guarantee. We therefore recommend a hybrid deployment in which LAC is the primary classifier (efficiency) but min-CPS sets are surfaced to clinicians as a secondary “proximity-band” display whenever the LAC set spans non-adjacent classes. This hybrid pairs naturally with the CORN ordinal-consistent classifier from Table 3 (lowest mean absolute ordinal error, MAOE 0.213, vs. multiclass CatBoost 0.264): in the proximity-first deployment we would substitute CORN for the underlying point classifier and run min-CPS on top, optimising both proximity loss and set-width simultaneously. We retain multiclass CatBoost + LAC as the headline pipeline in this paper because the binary and three-class targets dominate clinical-decision-support use cases, but the CORN+min-CPS hybrid is the right pipeline for proximity-first deployments and is implemented in the reproducibility package for community adoption.

Calibration-strategy sensitivity Two calibration strategies were compared as a sensitivity analysis: split conformal (50/50 calibration/evaluation split) and cross-conformal CV+ (5-fold rotation). CV+ achieved marginal coverage at or above the nominal target at all three CLs (80/90/95%), while split conformal dipped slightly below target at 90% CL for NSD-positive sub-staging and exhibited zero coverage on the Stage 4 minority class under the full-ordinal target ($n=3$ test patients, split conformal covers 0/3). CV+ recovered full Stage 4 coverage (3/3). Full per-method per-class coverage tables and the 2,201-patient jackknife⁺ [4] sensitivity check are reported in Supplementary S-2 (`supplementary_conformal_sensitivity.md`).

Feature Ablation

DaT-SPECT features are essential for binary NSD-ISS prediction (Table 5, visualised in Fig. 4): on the 21-feature strict-circularity primary, removing imaging+genetics reduces CatBoost AUC by 17.6 percentage points ($\Delta = -0.176$, paired bootstrap $p < 0.0001$). However, within the NSD-positive subgroup—conditioning on the subset of patients already confirmed as biologically positive—the delta is statistically indistinguishable from zero ($\Delta = +0.008$, $p = 0.21$), indicating that once biological positivity is established, clinical assessments are nearly sufficient for sub-staging. This conditional result is consistent with prior NSD-ISS validation

Table 5. Feature Ablation: Full vs. Clinical-Only Model (CatBoost AUC). **Top block:** 21-feature strict-circularity primary vs. 12-feature clinical-only common subset (the headline ablation visualised in Fig. 4). **Bottom block:** 22-feature reference vs. 12-feature, retained as audit trail.

Target	Full	Clinical (12)	Δ (pp)
<i>Primary specification (21 features, strict circularity, 5-fold CV)</i>			
Binary	0.901	0.725	−17.6
Three-class	0.897	0.797	−10.0
Full ordinal	0.915	0.823	−9.2
NSD+ subgroup	0.908	0.899	+0.8 (NS)
<i>Reference specification (22 features, audit trail)</i>			
Binary	0.979	0.727	−25.2
Three-class	0.944	0.797	−15.6
Full ordinal	0.954	0.823	−13.1
NSD+ subgroup	0.913	0.900	−1.4

Note: 21-feat primary block uses CatBoost-default 5-fold CV (random_state=42); deltas are paired-bootstrap (1,000 resamples) with 95% CI [−0.200, −0.155] for binary ($p<0.0001$) and [−0.005, +0.022] for NSD+ subgroup ($p=0.21$, statistically indistinguishable). 12-feat internal numbers from R4-Q6 like-for-like sweep (outputs/paper1_r2_responses/q_r4.q6_internal_12feat.json). 22-feat reference block retained for prior-art comparability; deltas are unpaired point estimates only.

Fig 5. Cohort-distribution domain shift driving external binary-classifier failure. UPDRS-III and UPDRS-I subscale distributions differ markedly between PPMI and BioFIND: **(a)** Bradykinesia, **(b)** Tremor, **(c)** Rigidity, **(d)** UPDRS-I Total. PPMI NSD-negative patients include healthy controls (low scores), while BioFIND S-negative patients are diagnosed PD (high scores), causing the binary classifier to learn a HC-vs-PD boundary rather than S+-vs-S−.

work [32,39] showing that the Simuni 2024 clinical sub-staging thresholds carry most of the sub-stage signal among confirmed NSD+ patients, and it is the basis for the two-stage deployment pattern we discuss in the Discussion.

Extended-feature sensitivity (pre-registered null) A pre-registered sensitivity analysis extended the 22-feature canonical schema to 33 features by adding six FreeSurfer cortical thickness measurements [18], four CSF biomarkers [26], and a Parkinson’s polygenic risk score [27] drawn from PPMI’s biomarker and imaging batteries. The SQL table assembly, cohort composition (all 2,201 patients retained; CTH 49.3%, CSF 36.8%, GRS 81.3% observed), and decision rule (halt migration if ≥ 3 of 4 CatBoost targets regress) were specified before execution. CatBoost balanced accuracy moved from 0.951/0.783/0.660/0.664 to 0.951/0.772/0.650/0.650 on binary / three-class / full-ordinal / NSD+ respectively, triggering the halt rule. All four deltas fall within the bootstrap 95% CI of the 22-feature baseline and the null of no improvement cannot be rejected. We attribute the non-improvement to (i) median imputation of partially-observed biomarkers, and (ii) DaT-SPECT already saturating the discrimination boundary (Table 5). Paper 1’s 22-feature schema is retained as canonical; richer imputation strategies that may recover marginal signal from CSF/CTH/GRS are deferred to future work. Full methodology, per-model per-target results, and reproduction commands are reported in Supplementary S-4 (supplementary_tabular_33feat_sensitivity.md).

External Validation: BioFIND

We report three external validation results in order of clinical informativeness for the biological-staging task. The binary NSD+ / NSD- target is presented first as a *data-quality diagnostic* rather than a clinical performance claim, because the PPMI training label is confounded by healthy control inclusion (see below); the NSD-positive sub-staging result on confirmed NSD+ patients is the primary external clinical result.

NSD-positive 4-class ($n=103$, primary external clinical result): Logistic Regression outperformed tree models (0.425 balanced accuracy, QWK 0.385, macro AUC 0.900), suggesting an approximately linear clinical-feature-to-stage mapping among confirmed NSD+ patients. Because the sub-staging task operates

Fig 6. Cross-model prediction disagreement on PDBP ($n=893$). Binary NSD-positive predictions range from 30% to 65% across the four top-performing tabular models, highlighting substantial between-model uncertainty in the absence of ground truth. Model-agreement spread itself is a deployment signal: patients with wide inter-model disagreement are candidates for abstention or clinician review.

entirely within the NSD+ population, it is structurally immune to the healthy-control confound that undermines the binary target, making it the most clinically meaningful external-validation signal in this study.

Three-class ($n=103$): AUC 0.703, suggesting moderate ranking ability despite poor calibration.

Binary ($n=108$, reported as a data-quality diagnostic): CatBoost achieved 0.516 balanced accuracy (AUC 0.637), near random chance. Root cause: PPMI NSD-negative class includes healthy controls with inherently low motor scores (UPDRS-III bradykinesia mean 7.37 in PPMI NSD-negative vs. 19.9 in BioFIND SAA-negative PD; both means computed on the common-feature external-validation cohort using the same pipeline as Table 3, and plotted in Fig. 5). Models therefore learned a healthy-vs-PD boundary rather than S+/S-. We report this number to surface the training-label confound rather than as a model performance claim; the pre-specified mitigation—retraining binary classifiers on PD-only subsets—is reported in the Results (Training-Label Confound).

External Predictions without Ground Truth: PDBP

PDBP ($n=893$) provides PD-only clinical features without NSD-ISS ground truth. Binary NSD+ predictions ranged from 30.2% (Random Forest) to 65.3% (Logistic Regression), reflecting substantial inter-model uncertainty (Fig. 6). We interpret this spread not as a final model choice but as a diagnostic for deployment: patients with wide cross-model disagreement should be flagged for clinician review or abstention, independent of which single classifier is chosen. The HBS cohort ($n=649$), which is missing 5 of the 12 common features, is reported in Supplementary the Supporting Information as a deployment cautionary tale (median-imputation of entirely-absent features produces uniformly NSD-negative predictions).

Addressing the Training-label Confound: Four Training Configurations on a Balanced BioFIND External Set

The Espay *et al.* [16] refutation of SAA-only NSD-ISS staging, together with the Simuni *et al.* [33] reply and the independent “Reconsidering the clinical foundations of NSD-ISS” [31] critique, converge on one methodological prescription: NSD-ISS prediction models must be evaluated on reference classes that exclude healthy controls, because mixing HC into NSD-negative conflates “not-PD” with “SAA-negative within PD.” Bentivoglio *et al.* [13] subsequently characterised the clinical/imaging phenotype of the SAA-negative PD subgroup, showing it is a real, identifiable PD sub-population rather than a data artefact. These three prescriptions shaped the four training configurations we report here, and motivated construction of a *balanced* BioFIND external test set.

Balanced BioFIND external set The published Russo 2025 [32] NSD-ISS BioFIND staging contains $n=103$ SAA-positive PD patients, all labelled NSD+. The BioFIND features release [14] contains $n=118$ PD patients: the additional 15 are SAA-negative PD patients, which Bentivoglio *et al.* [13] characterise as a distinct PD sub-phenotype and which map to NSD-negative under the NSD-ISS framework. We therefore construct a balanced BioFIND external set of $n=118$: 103 NSD+ and 15 NSD-negative, which to our knowledge is the first external NSD-ISS test set with both classes represented on a PD-only cohort.

Four training configurations Table 6 reports PPMI 5-fold cross-validation and balanced-BioFIND external performance for four training strategies.

Where the HC confound matters The magnitude of the training-label confound is feature-set dependent. With the *full 22-feature* set (Table 3), PD-only retraining yields essentially the same internal

Table 6. Training-strategy sweep: CatBoost binary classifier on the 12-feature common subset. PPMI values are 5-fold stratified CV (mean $\pm$ SD). BioFIND is the balanced external set ($n=118$: 103 NSD+ Russo-staged + 15 SAA-negative PD per Bentivoglio 2026 [13]).

Configuration	n_{tr}	PPMI (5-fold CV)		BioFIND (balanced external)	
		Bal. Acc.	AUC	Bal. Acc.	AUC
Baseline (full PPMI)	2,201	0.670 ± 0.026	0.738 ± 0.027	0.470	0.561
PD-only	1,747	0.618 ± 0.006	0.677 ± 0.016	0.521	0.561
Weighted (HC = SWEDD = 0.1)	2,201	0.652 ± 0.027	0.727 ± 0.026	0.497	0.524
Stage-A (HC vs. PD)	2,201	0.832 ± 0.020	0.931 ± 0.015	—	—

Note: PD-only keeps PD + Prodromal, drops HC + SWEDD. Weighted keeps all cohorts with HC and SWEDD sample-weights reduced to 0.1. Stage-A is the HC-vs-PD detection head of the hierarchical architecture discussed in the Discussion; no NSD-ISS external comparison is defined for this head.

result as the baseline (0.946 ± 0.014 balanced accuracy versus 0.951 ± 0.019 ; 0.982 ± 0.011 AUC versus 0.979 ± 0.013), because DaT-SPECT striatal binding ratio dominates the decision boundary and leaves no room for the classifier to exploit motor-score differences between HC and PD patients. The confound emerges only on the clinical-only 12-feature common subset used for external transportability, where the absence of imaging features lets the model fall back on the HC-vs-PD motor-score gap as a shortcut. On that subset, PD-only retraining changes the balanced-BioFIND point estimate of balanced accuracy from 0.470 to 0.521 (+5.1 percentage points on the point estimate) while reducing internal PPMI AUC from 0.738 to 0.677. A post-hoc 1,000-resample bootstrap on the $n=118$ balanced BioFIND set with the 15:103 class imbalance shows that 95% confidence intervals for the pre- and post-retraining balanced-accuracy estimates overlap substantially and straddle chance level; the +5.1 pp point improvement is therefore indicative of direction but is *not statistically significant* at $\alpha=0.05$ at the $n=118$ external sample size. We note that: the PD-only retraining removes a label-leakage shortcut in the PPMI internal decision boundary (and should therefore be preferred for transportability), but a definitive demonstration of external-cohort rescue requires a larger PD-only external test set than BioFIND currently provides. The clinical implication is clear: sites with DaT-SPECT access should use the robust full-feature model; sites without imaging should use the PD-only retrained 12-feature model for deployment. DaT-SPECT availability varies substantially across community-practice settings, where the imaging modality is FDA-approved but not universally reimbursed [35], so the two-model deployment protocol is essential for any realistic clinical uptake of NSD-ISS staging.

Intermediate weighting is not a substitute Sample-weighting HC and SWEDD at 10% (“Weighted” row) achieves an internal-CV result intermediate between full and PD-only (0.652) but yields worse external performance than PD-only in both balanced accuracy (0.497 vs. 0.521) and AUC (0.524 vs. 0.561). Full cohort exclusion is the effective mitigation; partial down-weighting leaks residual HC information through the kept mass.

A hierarchical alternative Stage-A—the HC-vs-PD/Prodromal detection head of a hierarchical architecture—achieves 0.832 balanced accuracy and 0.931 AUC on PPMI internal CV, confirming that the healthy-control-detection problem is easy and separable from NSD-ISS staging. For screening at sites where diagnostic status is unknown, a deployment pipeline can run Stage-A first and pass Stage-A-positive patients to the PD-only (Stage-B) NSD-ISS classifier; for deployment at specialty PD clinics where all patients are already diagnosed (as in BioFIND, PDBP, and most clinical deployment contexts) Stage-A can be bypassed and the Stage-B classifier applied directly. This two-stage architecture resolves the tension between the HC-inclusive training design of PPMI and the PD-only clinical deployment context that the Espay critique identifies. **End-to-end pipeline evaluation (R5-Q10):** chaining Stage-A (12-feature HC-vs-PD/Prodromal CatBoost, pooled OOF AUC 0.929) into Stage-B (21-feature primary CatBoost binary NSD-ISS, AUC 0.878 on the PD-clinic-like subset) under split-conformal abstention at 90% confidence on the $n=1,747$ PD/Prodromal cohort yields end-to-end coverage accuracy 0.850 at a 27.6% specialist-referral rate, with only 3.1% of true NSD+ patients incorrectly routed as healthy controls (cascading miss rate).

Tightening the Stage-A operating point to a 20% referral budget ($\tau=0.75$) holds coverage accuracy at 0.822 with a 13.4% cascading miss rate. The cumulative error is dominated by Stage-B abstention (22.3pp of the 27.6% budget) rather than Stage-A misclassification, which means the deployment-throughput bottleneck is the within-NSD+ multiclass uncertainty already documented in the conformal coverage analysis (the Results), not the upstream HC-vs-PD gate. Full operating-point sensitivity sweep ($\tau \in [0.30, 0.90]$) is at `outputs/paper1_r2.responses/q_r5_q10_end_to_end_pipeline.json` (SQL `run_id q_r5_q10_end_to_end`); model card, decision thresholds, calibration curves, and 5 example conformal outputs are in the deployment kit `DEPLOYMENT_KIT.md` (R5-Q9). **Temporal validation (R6-Q6 follow-up):** a prospective-style split that trains the cascade on PPMI 2010–2020 and evaluates on the late-wave PPMI 2021–2025 cohort ($n_{\text{test}}=790$) preserves overall coverage accuracy (0.851 vs. 0.850 random-CV baseline) but exhibits a $3\times$ increase in cascading miss rate (0.094 vs. 0.031), driven entirely by Stage-A drift on the late-wave cohort’s shifted prodromal/PD composition (86% vs. 80% in train). Stage-B is temporally stable ($\Delta\text{AUC} = -0.007$). The deployment implication is that prospective monitoring should focus on the HC-vs-PD detector (Stage-A) rather than the NSD-ISS staging model itself; quarterly Stage-A recalibration on a small labelled local cohort is the recommended drift-mitigation cadence (full operating-point sweep at `outputs/paper1_r2.responses/q_r6_q6_temporal_cascade.json`; SQL `run_id q_r6_q6_temporal_cascade`).

Discussion

Principal Findings

Our framework differs from prior PD / NSD-ISS prediction work (Supplementary Table S-2) in two ways: (i) it targets the NSD-ISS biological-stage label across four target formulations jointly, and (ii) it pairs that benchmark with distribution-free uncertainty quantification—the combination is, to our knowledge, novel for the NSD-ISS framework. Lian *et al.* [1] targeted clinical milestones on PPMI ($n=884$), Russo *et al.* [32] published a descriptive NSD-ISS staging on BioFIND ($n=103$), Simuni *et al.* [40] reported transition Kaplan–Meier estimates on PPMI ($n=995$), and Diaz-Rincon *et al.* [12] applied conformal prediction to PD medication-need assessment ($n=427$); none attempted multi-target NSD-ISS prediction and none combined cross-conformal uncertainty with multi-cohort external validation.

This study provides a strict-circularity machine-learning benchmark for NSD-ISS biological stage prediction (the first published, to our knowledge), yielding three findings with direct clinical implications. First, under the strict-exclusion specification that removes every staging-algorithm variable from the predictor set (21 features; see the Methods (Circularity Audit)), binary NSD+ AUC is 0.901 [0.887, 0.915]—a paired-bootstrap ablation against the 12-feature clinical-only subset reduces this to 0.726 [0.705, 0.748] ($\Delta = -0.176$, 95% CI $[-0.200, -0.155]$, $p < 0.0001$), confirming that caudate-region dopaminergic imaging and genetic carrier status are essential for biological detection while non-staging clinical variables alone are insufficient. Re-including the caudate/putamen ratio in the predictor set inflates binary AUC by +0.077 and three-class macro-AUC by +0.047 (Table 3, reference vs. primary rows); the magnitude of this inflation is why the ratio is excluded, and underscores the NSD-ISS framework’s vulnerability to second-order D-anchor leakage through derived imaging features. This aligns with the NSD-ISS validation evidence that DaT-SPECT striatal binding is the dominant time-to-clinical-milestone predictor [39] and with the established diagnostic role of dopaminergic imaging in PD differential diagnosis [35], while making explicit the boundary between staging-variable leakage and caudate-region residual signal.

Second, the 21-feature strict-exclusion CatBoost on NSD+ sub-staging achieves AUC 0.908 [0.889, 0.926]; the 12-feature clinical-only subset reaches 0.899 [0.874, 0.923] on the same target. A paired-bootstrap test of the difference yields $\Delta = +0.008$ (95% CI $[-0.005, +0.022]$, $p = 0.21$, $n = 779$)—the two specifications are statistically indistinguishable on the sub-staging task, while the same 21-to-12-feature reduction on the binary target collapses AUC from 0.901 to 0.726 ($\Delta = -0.176$, 95% CI $[-0.200, -0.155]$, $p < 0.0001$). This asymmetric ablation is the sharpest empirical endorsement in this paper of the two-stage deployment pattern: DaT-SPECT is essential once, for the initial NSD+ vs. NSD– decision; after biological confirmation, clinical-only variables suffice for NSD+ sub-staging with no statistically detectable loss.

Modality-robustness consolidation under temporal shift (Q.E.D. OPTION 1): a head-to-head

Table 7. Deployment-personas executive summary. Five canonical clinical deployment scenarios for the NSD-ISS predictive framework, with the recommended model variant, headline performance, and uncertainty-aware action policy. “Stage-A” = HC-vs-PD/Prodromal detector (12-feat clinical, AUC 0.929); “Stage-B” = within-PD NSD-ISS classifier (21-feat or 12-feat depending on imaging access).

Persona		Recommended model		Headline metric		Action policy
Specialty clinic (DaT+SAA)	PD with imaging	Stage-B 21-feat primary, CatBoost + CV+ LAC		NSD+ 0.908 [0.889, 0.926]	AUC	singleton→report; multi→present alternatives; empty→repeat assessment
Community clinic (no imaging / SAA)	PD	Stage-B 12-feat clinical-only, CatBoost + CV+ LAC		NSD+ 0.899 [0.874, 0.923] (R4-Q6, R5-Q3 striatal-free +0.5 pp)	AUC	same as above; clinical-only sub-staging is the imaging-agnostic variant (R5-Q3)
Trial-enrichment screening (HC+PD cohort)	mixed	Stage-A → Stage-B cascade with formal abstention at 90% CL		End-to-end accuracy 0.850, referral 27.6%, cascading miss 3.1% (R5-Q10)		route Stage-A-positive patients to Stage-B; route abstentions to specialist (DEPLOY-MENT_KIT.md)
Cross-site deployment (multiclass NSD-ISS)		Stage-B + Mondrian CP recalibrated on $n_{\text{cal}} \geq 40$ –50 labelled local patients (R6-Q2)		per-class coverage 0.131 → 0.944 (three-class), 0.356 → 0.876 (NSD+)		site-specific Mondrian recalibration before reporting probabilities
Drift monitoring (post-deployment)		Recommended: quarterly Stage-A recalibration on small labelled local cohort (derived from R6-Q6)		Stage-B temporally stable ($\Delta\text{AUC} = -0.007$); Stage-A drift drives 3× cascading miss		Suggested: monitor HC-vs-PD detector; refresh Stage-A on ≥ 50 labelled local cases per quarter (cadence is a derived recommendation, not separately validated — see DEPLOY-MENT_KIT.md §8)

between 21-feat and 12-feat NSD+ sub-staging under three CV regimes—standard 5-fold stratified, enrollment-wave GroupKFold (3 folds $\times$ 5 seeds), and pre-2014 vs. 2014+ era-LOCO—reveals that under both temporal regimes the 12-feat clinical-only model *outperforms* the 21-feat by 4.0–5.2 pp ($\Delta = -0.052$ wave-GroupKFold; $\Delta = -0.040$ era-LOCO), reversing the standard-CV equivalence ($\Delta = -0.001$). The directional flip is consistent with caudate SBR + caudate asymmetry features carrying scanner/era-specific signal that does not generalise forward, while UPDRS / MoCA / RBD / ESS clinical features are temporally invariant. The deployment implication strengthens beyond “acceptable fallback”: clinical-only NSD+ sub-staging is *preferred* for prospective cross-era deployment, not merely tolerated. Full results at `outputs/paper1_r2.responses/q_gap_2.nsd.plus.modality.consolidation.json`. Operationally, the clinical-only sub-staging model at AUC 0.899 is the **imaging-agnostic Stage-B variant requested by R7-Q9**: it can support trial stratification and recruitment screening at community-practice sites that lack

DaT-SPECT or SAA access, extending NSD-ISS-based trial enrichment beyond the specialised imaging centres where the framework was originally validated. The R5-Q3 striatal-free upper-bound sensitivity (Δ NSD+ AUC +0.5 pp under removal of all caudate features in addition to the already-excluded putamen + ratio + NP3TOT + NP1COG) confirms that within-NSD+ sub-staging is genuinely independent of imaging-anchor information; an imaging-agnostic deployment is therefore not merely a fallback for resource-limited sites but a principled clinical pipeline backed by the residualization-robust NSD+ result. A complementary rule-based Simuni-threshold baseline applied to the same task recovers the labels by construction (the Methods (Circularity Audit)); the ML value proposition is therefore not to out-predict the rule when the rule-defining variables are present, but to generalise to cohorts with partial missingness of those variables—the realistic community-deployment regime.

Third, the PPMI-to-BioFIND external validation of the binary target (balanced accuracy 0.516) exposed a training-label confound rather than a model limitation: PPMI’s NSD-negative class mixes healthy controls with prodromal patients, so the binary classifier trained on the full PPMI cohort learned a healthy-versus-PD boundary rather than a within-PD SAA-positive-versus-SAA-negative boundary. Retraining the same CatBoost classifier on the PD-only subset of PPMI (the Results (Training-Label Confound)) corrects this confound at source and demonstrates that the PD-within-PD discrimination is recoverable from the same 12-feature common set. Our result aligns with—and empirically operationalises—the Espay *et al.* [15] critique that NSD-ISS prediction models must explicitly account for the composition of their NSD-negative reference class, since mixing healthy controls with medication-treated prodromal patients introduces two confounds (dopaminergic treatment status and disease status) into a putatively biological label.

Tabular vs. Graph-Based Models

Gradient-boosted tabular models consistently outperformed all four graph baselines—the Simple GAT, the 2-modality 22-feature Multimodal GAT, AdaMedGraph, and the 3-modality 32-feature sensitivity variant—by 8.1–33.7 percentage points in balanced accuracy across the four targets (min: binary AdaMedGraph at -0.081 balanced accuracy; max: NSD-positive AdaMedGraph at -0.337 ; see Table 3 (“Graph-based” rows)). The Multimodal GAT with GATConv and cross-modal attention improved over a simple single-stream GAT (particularly for multiclass: three-class 0.705 vs. 0.666), but the k -NN patient-similarity graph did not provide information beyond what tree ensembles capture directly from feature vectors. AdaMedGraph is the only graph baseline that beats the Multimodal GAT on binary classification (0.870 vs. 0.825), but it collapses on ordinal targets where per-feature-graph ensembling cannot recover from minority-class sample size (Stage 4 $n=17$). Within the tabular family, the four-way nested-HPO + TabPFN + AutoGluon comparison in Table 3 (“Tabular SOTA” rows) shows that CatBoost, LightGBM, TabPFN v2 (no HPO), and AutoGluon full-pool ensembling achieve statistically indistinguishable AUC across all four NSD-ISS targets, with 95% bootstrap CIs overlapping on every target (nominal winner is TabPFN on 3 of 4 by point estimate, but no pairwise difference reaches $\alpha=0.05$). This empirically validates the Zabérgja 2024 [46] + Hollmann 2025 [49] + Erickson 2020 [50] prediction of tabular-architecture convergence at the $n \approx 2,000$ clinical-cohort regime, and refines the Grinsztajn–Shwartz–Ziv narrative: trees do not lose to deep or automated methods at this n , but they no longer dominate them either. Our CatBoost choice is therefore deployment-motivated (fewer dependencies, faster training, native missing-value handling) rather than discrimination-motivated; practitioners with different deployment constraints (zero-tuning preference $\rightarrow$ TabPFN cloud; ensemble robustness $\rightarrow$ AutoGluon; local speed-over-all $\rightarrow$ LightGBM) can substitute another member of this Pareto-indifferent family without sacrificing discrimination.

A pre-registered sensitivity variant with three modality streams and 10 further observed PPMI features (6 cortical-thickness regions per Fischl [18] and 4 CSF biomarkers per Mollenhauer *et al.* [26]) was evaluated to isolate whether the 2-modality gap is architectural or feature-count-limited (Supplementary S-3). The 3-modality 32-feature Multimodal GAT (Table 3 (“Graph-based” rows), fourth row per target) closes the CatBoost gap to -1.7 pp on binary (bal. acc. 0.934, AUC 0.978) and -3.0 pp on three-class (bal. acc. 0.753, AUC 0.924), but widens it on full ordinal (-15.6 pp) and NSD-positive subgroup (-25.4 pp). The two targets where the variant improves are the clinically most important (screening and triage); the two where it degrades are the rarer-class ordinal tasks where the Extended modality’s $\sim 50\%$ cortical-thickness and $\sim 35\%$ CSF coverage introduces median-imputation noise into minority classes (Stage 4 $n=17$). We interpret this as

partial support for both readings: Grinsztajn 2022 holds on average, but a graph-attention architecture given sufficient observed biological features is genuinely competitive with tree ensembles on the two clinically valuable targets. Graph models may also become advantageous for inductive transfer to cohorts with incomplete features, where graph-informed imputation can leverage similar patients; evaluating that hypothesis on NSD-ISS staging is left as future work.

Conformal Prediction for Clinical Deployment

Conformal prediction performance is summarised in the Results and Fig. 3; the deployment-relevant point is that empty-set abstentions (mean $|C| < 1$) are a feature, not a defect—they surface exactly the patients on whom additional clinical workup would have the highest information value (the Supporting Information). Per-stage rare-class behaviour is decomposed in Supplementary the Supporting Information: cross-conformal CV+ achieves marginal coverage 1.000 on Stage 4 ($n=17$) by widening the prediction set, but the underlying argmax discriminator’s F1 of 0.182 reflects the fundamental noise floor at $n=17$ training examples—a structural cheap pass for the conformal guarantee that should be communicated transparently to deployers.

Confounder sensitivity Five pre-specified sensitivity analyses directly pre-empt the question of whether the 21-feature primary binary AUC (0.901) is age-, sex-, scanner-, site-, or recruitment-era-confounded; full results are given in Supplementary S-5. All deltas below are reported against the 21-feat primary unless otherwise stated; a separate 22-feat-reference re-run confirms the same direction and magnitude (all deltas within 0.005 AUC).

First, a 1:1 greedy age-matched cohort ($n=1,558$, caliper ± 2 years $\approx 0.2 \times \text{SD}$ per the canonical matching-caliper recommendation [3], residual matched age $\Delta = -0.002$ years vs. the full-cohort $+0.576$ -year gap) yields binary CatBoost AUC 0.876 [95% CI 0.857–0.892] on the 21-feature primary specification, within 0.025 of the 0.901 full-cohort primary and below the pre-registered 0.03 confound threshold; the same analysis on the 22-feature reference specification yields 0.969 [0.960–0.978], confirming the conclusion is not specification-dependent. This is consistent with the published finding that age and sex jointly explain $< 10\%$ of DaT-SPECT SBR between-subjects variance versus the $\sim 50\%$ reduction that defines pathological DaT loss [17, 34, 42].

Second, sex-stratified CatBoost on the 21-feature primary yields nearly identical male and female binary AUCs (bootstrap interaction $\Delta = +0.0029$, 95% CI $[-0.025, +0.031]$, $p=0.822$), consistent with the published age/sex-correction analysis [34]; the same conclusion holds on the 22-feature reference specification ($\Delta = +0.0004$, $p=0.914$).

Third, a PPMI enrollment-wave leave-one-cohort-out analysis (early 2010–2013, middle 2014–2020, late 2021–2025 reflecting the PPMI 1.0 $\rightarrow$ 2.0 transition) yields per-wave binary AUCs in $[0.871, 0.891]$ on the 21-feature primary (SD across waves ≈ 0.010)—a tighter range than the 22-feature reference specification’s $[0.947, 0.992]$ because the primary excludes the ratio feature that amplifies wave-era imaging drift. The classifier generalises across PPMI eras without exceeding clinically acceptable variance on either specification. **R7-Q4 wave-grouped repeated CV (full cohort).** A follow-up reframes the single-shuffle wave-LOCO as 3-fold GroupKFold (1 fold per wave) $\times$ 5 shuffle seeds (15 fold-AUCs total) on the full $n=1,845$ cohort with non-null enrollment date. Mean AUC 0.879 ± 0.008 across the 15 folds, reproducing the wave-LOCO range with $6 \times$ tighter variance estimate. Per-wave: early 2010–2013 0.890 ± 0.001 , middle 2014–2020 0.872 ± 0.004 , late 2021–2025 0.875 ± 0.002 . Unconditional wave-ICC = 0.082 (linear-scale MixedLM) places enrollment wave in the same “meaningful confounder” band (0.05–0.20) as the R5-Q5 site-ICC of 0.059; conditional ICC failed to converge once the 19 patient predictors are held fixed, indicating the predictors absorb essentially all wave-attributable variance. *Verdict:* discrimination is invariant across PPMI eras (mean 0.879; the 0.85 threshold is an internally-defined deployment heuristic, not a literature-validated standard), but raw between-wave variance share (8.2%) motivates era-aware recalibration for prospective deployment (outputs/paper1_r2_responses/q_r7_q4_wave_grouped_cv.json; SQL run_id q_r7_q4_wave_grouped_cv).

Fourth, a pre-registered DaT-SPECT protocol leave-one-cohort-out analysis was run on the 2,137-patient analyzed-baseline-scan cohort (protocol 001 $N=965$ vs. 002 $N=1,141$; edge training bucket 004 + T011). On the 21-feature primary, per-protocol binary AUCs are 0.903 (hold out 001) and 0.912 (hold out 002), cross-protocol mean 0.907 ± 0.006 —a tighter SD than the 22-feature reference (0.978 ± 0.016), indicating

that the primary specification is better-calibrated across acquisition protocols. This partially disentangles the scanner/reconstruction-era effect from the cohort-recruitment-era shift absorbed in the enrollment-wave analysis.

Fifth, a pre-registered site leave-one-site-out (site-LOSO) analysis was run on the T1-MRI imaging-covered subsample ($n=647$ of 2,201, 29.4%). Site identifiers were extracted from LONI IDA PPML*.xml metadata and persisted as SQL. This yields 14 per-site folds with $n \geq 20$ plus one pooled-small-sites fold ($n=209$). Decision rule was locked before any result was inspected (PASS if pooled per-site AUC SD $\leq 3 \times$ protocol-LOCO SD AND min per-site AUC ≥ 0.90). Site-LOSO FAILED the pre-registered decision rule on both specifications: 22-feat reference (pooled mean 0.955 ± 0.070 , min 0.735 at site 290) and 21-feat primary (pooled mean 0.832 ± 0.091 , min 0.500). The failure was driven by two small folds with extreme class imbalance (site 290: $n=21$, 81% NSD+; site 007: $n=27$, 85% NSD+)—a regime in which AUC is known to be high-variance [43]. We note two important caveats. (i) *Subsample selection bias*: the 647-patient MRI-covered subsample is not a random draw from the full 2,201 cohort. T1-MRI acquisition was more common at larger academic centres with richer imaging infrastructure, biasing the site-LOSO evaluation toward the higher-resource segment of the PPMI site network and away from the smaller community-feeder sites where deployment robustness matters most. (ii) *Site-identifier recovery is structurally blocked at the DUA level*: an attempt to recover site identifiers for the remaining 1,554 patients from the PPMI harmonised release failed because the Site Number (CNO) column is explicitly suppressed under the PPMI Data Use Agreement (the `site_aprv` field present in the harmonised release is a site-approval date, not a site identifier, and confirms the suppression). The recovery pathway available to us is per-modality LONI IDA XML metadata extraction: the T1-MRI loader at `scripts/load_paper1_site_assignments_full.py` (committed 2026-04-24) is generalisable to DaT-SPECT XMLs (which carry the same `siteKey` field), and the corresponding 2,137-patient DaT-SPECT site recovery is in our R3 follow-up plan. We did not include it in the current submission for two reasons: site identifiers in DaT-SPECT XMLs are technically present but their cross-modality consistency with T1-MRI siteKeys requires manual verification (each modality has its own LONI collection-level metadata schema), and we wanted to avoid a partial recovery that overrepresented imaging-rich sites in a different way than the T1-MRI subsample already does. We therefore present the current MRI-cohort site-LOSO as honestly site-restricted; the full-cohort site-LOSO is the natural R3 follow-on and is already engineering-ready. The recommended deployment pattern is on-site calibration prior to clinical use, with protocol-LOCO (Analysis IV) serving as the cleaner cross-scanner generalisation evidence. **R6-Q4 failure-mode decomposition**: per-site analysis identifies the failure as concentrated on small ($n < 25$) and class-imbalanced ($> 70\%$ one class) sites: the three worst-AUC folds are site 290 ($n=21$, 81% NSD+, AUC 0.662), site 096 ($n=21$, 48% NSD+, AUC 0.700), and site 088 ($n=22$, 73% NSD+, AUC 0.729); small-site mean AUC is 0.796 vs. big-site 0.873 ($\Delta = +0.077$). The failure is class-symmetric (mean Δ recall NSD+ minus NSD− across folds = $+0.013$), not a single-class collapse, ruling out a class-imbalanced-classifier failure mode. A pre-registered SMOTE-oversampling sensitivity (excluded from the primary protocol per the conformal-exchangeability concern in the Discussion (Limitations)) yields $\Delta\text{AUC} = -0.017$, confirming targeted resampling does not recover the failed regime within pre-registration constraints. **R5-Q5 grouped-repeated-CV reframe**: reframing the failed strict site-LOSO as 5-fold GroupKFold with site as the grouping variable $\times$ 5 shuffle seeds (25 fold-AUCs total) reduces the small-fold imbalance pathology (SD $0.091 \rightarrow 0.050$, -45% ; min lifts from $0.500 \rightarrow 0.748$) but does *not* rescue the absolute level: mean grouped-CV AUC 0.817 trails the 0.85 deployment-readiness threshold, and the unconditional site intra-class correlation ICC=0.059 (linear-scale MixedLM) places PPMI site in the “meaningful confounder” band (0.05–0.20). The softer methodology therefore confirms the strict-LOSO failure was *not* a methodological artifact—site is a genuine source of variance that prospective external-site validation will need to address before deployment (outputs/paper1_r2_responses/q_r5.q5_site_grouped_cv.json; SQL run_id q_r5.q5_site_grouped_cv).

Disease-progression proxy interaction (R2-Q9 follow-up) An exploratory follow-up requested by Reviewer 2 stratified the cohort by tertiles of an UPDRS-III bradykinesia+rigidity progression proxy. Bootstrap interaction tests with BH-FDR correction find a statistically significant target $\times$ progression-tertile interaction on binary (max $|\Delta| = 0.037$, exceeds the pre-registered 0.03 threshold) and three-class (max $|\Delta| = 0.098$, $p_{\text{FDR}} < 0.001$). We interpret this as an *expected diagnostic confound* rather than a fairness bias:

higher-progression patients cluster in Stages 3 and 4, where discriminating adjacent late stages is intrinsically harder regardless of the classifier. A cleaner disease-duration variable than a UPDRS-III severity proxy would disentangle the confound from genuine progression-stratified accuracy; we flag this as future work. Full per-target stratified AUC table at `outputs/paper1_r2_responses/q9_extended_subgroup.json`.

Internal Versus External Deployment Calibration

Calibration results (Fig. 7) reveal a pattern that dominates external deployment considerations for any NSD-ISS model. Internal CatBoost expected calibration error (ECE) on PPMI is excellent across all four target formulations—0.018 binary (post temperature scaling on the per-fold OOF predictions, vs. 0.053 raw pre-scaling reported in the abstract), 0.012 three-class, 0.035 full-ordinal, and 0.028 NSD+ subgroup (Supplementary S-2)—well below the 0.05 threshold typical of well-calibrated clinical models [53]. External calibration on BioFIND, however, degrades by an order of magnitude: ECE 0.399 binary, 0.301 three-class, 0.278 NSD+; Hosmer–Lemeshow $p=0$ on binary [54]. Logistic regression, despite achieving lower internal AUC than CatBoost, delivers meaningfully better external NSD+ calibration (ECE=0.229 vs. CatBoost’s 0.278), confirming the classical finding that simpler linear models transport better under distribution shift [55]. Combined with the external conformal coverage numbers in the Results (binary: 0.915; three-class: 0.499; NSD+: 0.707) this gives a coherent picture: the binary NSD-positive decision boundary transports; within-NSD+ multiclass sub-staging does not transport without recalibration on a target cohort. Practically, this means deployment at a new site should (i) run temperature scaling [53] on a small local calibration set before reporting probabilities, (ii) prefer binary NSD+ detection plus clinical-only NSD+ sub-staging (the two regimes where calibration transports best) over full-ordinal reporting, and (iii) treat three-class external predictions as screening triage only unless the deploying site has recalibrated locally. This is the strongest methodological recommendation of this paper for prospective external use.

ComBat protocol harmonization (R6-Q3 follow-up): negative result A pre-registered sensitivity tested whether empirical-Bayes ComBat harmonization (Wakasugi 2024) of the four caudate DaT-SPECT features by PPMI acquisition protocol ($n=2,137$; protocols 001/002/edge), preserving AGE/SEX/NSD-ISS-stage as covariates, would improve external transportability. The internal CV result is a clean DEGRADATION: ComBat reduces 21-feature CatBoost AUC by 0.14–5.65 percentage points across the four NSD-ISS targets (binary -5.65 , three-class -4.23 , full-ordinal -3.69 , NSD+ -0.14 pp; the small NSD+ delta is consistent with the R3-Q3 finding that within-NSD+ sub-staging is dispensable from striatal information). Mechanism: PPMI protocols 001 and 002 enrolled stage-different patient cohorts (the protocol-vs-stage marginal is non-orthogonal), so the empirical-Bayes shrinkage on protocol means strips biologically meaningful signal even when stage is included as a categorical covariate—violating ComBat’s covariate-preservation assumption (Johnson 2007 / Fortin 2017). Cross-protocol LOCO standard deviation does not shrink under ComBat (mean $\Delta SD = -0.0013$). External BioFIND transportability is unmeasurable on this analysis because the AMP-PD v4 BioFIND extract carries clinical features only (no DaT-SPECT). The ComBat negative result is consistent with the broader Q1 finding (the Results (Tabular Models)) that PPMI-trained calibrators cannot fix BioFIND prevalence shift; tree-based learners on raw SBR already absorb the protocol heterogeneity that ComBat is designed to remove. Full per-target results at `outputs/paper1_r2_responses/q_r6_q3_combat_harmonization.json` (SQL run_id `q_r6_q3_combat`, 8 rows).

Domain-shift mitigation beyond temperature scaling Temperature scaling is a necessary first step—it rescales a classifier’s softmax without changing its decision boundary and is therefore safe to apply on a small target-site calibration set—but it does not by itself correct the covariate and prior shifts that drive the order-of-magnitude ECE degradation observed here. Three additional mitigations are standard for this class of shift:

- (i) *ComBat-style empirical-Bayes batch correction* [44] removes site- and scanner-specific additive/multiplicative effects from imaging features while preserving biological variation. For the 12-feature clinical-only subset deployed across cohorts ComBat is inapplicable (features are not

Fig 7. Internal vs. external calibration of CatBoost across four NSD-ISS target formulations. (a) Expected calibration error (ECE): internal PPMI calibration is excellent across all targets (0.012–0.035, below the 0.05 well-calibrated threshold), while external BioFIND ECE degrades by roughly an order of magnitude (0.278–0.399). (b) Brier score shows the same degradation pattern. Practical implication: deployment at a new site requires local-cohort recalibration before probability scores can be reported; binary plus clinical-only NSD+ sub-staging are the regimes where recalibration is most feasible.

imaging-derived), but it is the recommended pre-processing step for any future multi-site PPMI extension.

- (ii) *Density-ratio importance weighting* reweights each training example by the ratio of target-to-source density over a small set of shift-informative covariates (age, UPDRS-III total, disease duration), correcting for systematic cohort-composition differences without labelled target data. The PD-only retraining result in the Results (Training-Label Confound) is the simplest special case, implementing the density ratio as a hard 0/1 cohort-inclusion gate.
- (iii) *Saerens-style prior-probability-shift correction* adjusts output probabilities by the ratio of target-prior to source-prior, assuming invariant class-conditional features. This is directly relevant to BioFIND’s 95.4% SAA+ prevalence vs. PPMI’s 3.0%.

Deployers with a modest labelled target calibration set ($n \gtrsim 50$) should layer these three corrections on top of temperature scaling. We do not apply them in the present paper’s primary results because the 103-patient labelled BioFIND external set produces unstable density-ratio estimates; their absence is why the external-deployment figures in the Results should be read as lower bounds, rather than ceilings, for sites that can budget a local calibration cohort.

Limitations

Observational study design (NSD-ISS data-availability constraint) The NSD-ISS framework was published in January 2024 [38]; no prospectively-NSD-ISS-staged cohort currently exists or could exist for this study. PPMI (NCT01141023 [24]) is the canonical longitudinal NSD-ISS-staged cohort and is observational by design; BioFIND [32] is also observational. Retrospective application of the NSD-ISS staging algorithm to existing cohorts (as we and Russo *et al.* do) is therefore the only feasible evaluation strategy at the time of writing. A truly prospective NSD-ISS validation requires future MJFF-funded cohort enrollment with NSD-ISS staging built into the protocol from baseline, which is outside the scope of any single paper. The five pre-registered confounder analyses (age, sex, enrollment wave, DaT-SPECT protocol, site-LOSO) directly characterise the residual selection-bias risks introduced by the observational design; the honest site-LOSO failure (the Results) is reported transparently rather than hidden.

Several additional limitations warrant explicit discussion. The most significant is severe class imbalance: Stage 4 comprises only 17 patients (0.8% of the cohort), which limits reliable evaluation of minority-class performance and inflates variance in per-class metrics. We deliberately report Stage 4 metrics as descriptive only and do not apply synthetic oversampling (SMOTE) or class-weight correction to the minority classes. Synthetic oversampling would violate the exchangeability assumption that underwrites the cross-conformal coverage guarantee—calibration-set exchangeability requires that the calibration fold be drawn from the same distribution as deployment patients [43], which synthetic Stage 4 samples are not—and stratified CV with at least three Stage 4 patients per fold already delivers the most statistically defensible assessment the sample size permits. Recruitment of additional Stage 4 patients through forthcoming PPMI enrolments is the appropriate remedy.

Construct validity of NSD-ISS labels under sparse SAA coverage (R2-W11) SAA coverage in PPMI is limited to 12.6% of participants (277/2,201), so the S anchor relies on D-anchor status for the remaining 87.4%; we follow the Simuni 2024 convention of assigning patients with D+ status (by DaT-SPECT) and absent SAA to the prodromal NSD+ side of the dichotomy only when at least one additional prodromal criterion is present, and mark the remainder as unclassified (full decision-path counts

Fig 8. Subgroup fairness by genetic-carrier status (21-feature primary, binary NSD-ISS).

Per-genotype subgroup AUC with 95% bootstrap CIs on the 21-feature strict-circularity primary.

Mutually-exclusive carrier strata: LRRK2+ (any), $n=175$, AUC 0.867 [0.811, 0.918]; GBA+ only (not LRRK2+), $n=104$, AUC 0.942 [0.890, 0.982]; APOE- $\epsilon 4$ + only (not LRRK2+, not GBA+), $n=375$, AUC 0.876 [0.839, 0.910]; Non-carrier, $n=1,547$, AUC 0.901 [0.886, 0.917]. All four carrier-subgroup CIs overlap the non-carrier reference band (blue shaded), indicating no systematic performance disparity by genetic-carrier status on the binary endpoint. **TreeSHAP feature attribution (R6-Q5):** cross-model SHAP analysis validates the imaging/clinical modality complementarity that motivates the two-stage deployment. The 21-feature binary primary's top-3 SHAP features are caudate DaT-SBR variants—CAUDATE_MEAN_SBR (mean |SHAP| 1.40), CAUDATE_R_SBR (0.68), CAUDATE_L_SBR (0.49)—confirming that binary NSD+ detection is dopaminergic-anchor-driven (consistent with the residualization sensitivity of the Results (Tabular Models) R3-Q3 and the striatal-free upper bound). The 12-feature clinical-only NSD+ sub-staging model's top-3 features are UPDRS-III motor subscales—UPDRS3_BRADYKINESIA (0.46), UPDRS3_AXIAL (0.45), UPDRS3_RIGIDITY (0.37)—confirming that within-NSD+ sub-staging is driven by motor severity, not imaging. The two stages therefore use distinct, biologically coherent feature axes (Supplementary Fig. R6-Q5; full top-10 + per-class breakdowns at `outputs/paper1_r2_responses/q_r6_q5_shap_analysis.json`). TreeSHAP attributions under correlated features (caudate-mean and caudate-L/R SBR; partial $r = 0.85$ caudate $\leftrightarrow$ putamen per R3-Q3) carry the standard interpretability caveat. **Extended fairness audit (R5-Q6):** a follow-up extends this analysis to age tertiles + sex on *both* the 21-feature binary model and the 12-feature clinical-only NSD+ sub-staging model (the deployment-relevant pipeline of the Results (Feature Ablation)). Of 6 axes (sex, age tertile, carrier $\times$ 2 models), 4 PASS at the pre-registered $|\Delta\text{AUC}| < 0.03$ threshold; the 2 failures are both on the LRRK2+ stratum— $\Delta = 0.046$ (binary, $n=175$) and $\Delta = 0.049$ (NSD+, $n=104$)—a $\sim 5\text{pp}$ gap consistent across both deployment models and consistent with the documented distinct neurodegeneration trajectory of LRRK2 carriers. The deployment recommendation is therefore targeted LRRK2+ recalibration before clinical use; sex and age tertile transport without recalibration on both models. **R6-Q8 finer-grain extension:** a 10-decile age stratification on both models retains AUC ≥ 0.86 across every decile (9/10 PASS at $|\Delta\text{AUC}| < 0.03$ on each model; Spearman age-AUC trend $p \geq 0.28$), confirming no monotonic age-performance gradient internally. External BioFIND sex stratification is consistent with internal ($\Delta\text{AUC} = -0.003$ for evaluable Female stratum; Male is single-class on BioFIND); external age tertile stratification is degenerate due to the cohort's 95.4% NSD+ uniform composition (per-tertile $n=32\text{--}39$ with 1–3 negatives), reflecting cohort design rather than a model failure mode. Full per-axis interaction tests, calibration-per-subgroup ECE, and the 6-panel forest plot are at `outputs/paper1_r2_responses/q_r5_q6_fairness_extended.json` (SQL run_id `q_r5_q6_fairness`, 18 rows; R6-Q8 extension at `q_r6_q8_subgroup_extended.json` adds 25 rows for age deciles + external).

in Supplementary the Supporting Information). A construct-validity concern arises that the resulting NSD-ISS labels predominantly reflect D-anchor signal rather than the integrated S+D biological phenotype the framework intends. We address this concern directly using the externally-validated SAA prediction model of Venuto *et al.* [65] (medRxiv 2025), which derives SAA+ probability from non-invasive features (UPSIT age- and sex-specific percentile, sex, constipation history, LRRK2/GBA carrier status) with PPMI-internal AUROC 0.920 and S4-external AUROC 0.976 (Fig. 9). Applying Venuto's published coefficients off-the-shelf (no PPMI re-fitting; LRRK2 sub-variants approximated as 90/10 G2019S/I2020T:R1441G/C per Venuto Table 1) to our SAA-tested cohort yields AUROC 0.810 on the 138 patients with both SAA and UPSIT—short of the 0.920 reference, attributable to the absent LRRK2 sub-variant detail and the SCOPA-AUT5 thresholding difference, but well above chance. Imputing SAA status for the **647 NSD-positive PD patients lacking SAA** (the cohort to which the construct-validity concern directly applies), Venuto's predictor estimates a **91.5% S+ rate** (236/258 imputable with UPSIT)—squarely between our SAA-tested NSD+ subset's observed 77.3% S+ rate, the Siderow *et al.* (2023) [67] PPMI-wide $\sim 88\%$, and Venuto's reported 93% in sporadic PD. The implication is that NSD-ISS labels assigned via the SAA-missing-D-positive decision path are biologically concordant with what an externally-validated S-anchor predictor would assign—the model is learning a dual-anchor-consistent

Fig 9. S-anchor coverage and Venuto-2025 construct-validity check. (a) SAA observation coverage stratified by NSD-ISS stage: 1,273 of 1,418 Stage-0 patients (89.8%) and 647 of 779 NSD-positive PD patients (83.1%) lack SAA testing. (b) NSD-positive S-anchor positivity rates: our SAA-tested NSD+ subset shows 77% S+; Venuto-2025-imputed S+ on the 258 SAA-missing-with-UPSIT NSD+ subset is 91%, sitting in the literature band (Siderowf 2023 [67] 88%, Venuto 2025 [65] sporadic-PD 93%). The convergence of imputed and literature S+ rates supports construct validity of the D-anchor-driven NSD-ISS labels.

biological signal rather than free-floating algorithmic recapitulation. We retain the original D-anchor-driven labels as primary because Venuto’s confidence intervals are not yet PPMI-recalibrated for this specific application, and present the imputation as a supplementary construct-validity check (full results at `outputs/paper1_r2_responses/q_gap_a_venuto_saa.json`). Broader SAA screening in future PPMI releases, and complementary digital risk scores [66] that correlate independently with both DaT and SAA biomarkers, will allow more direct sensitivity analyses.

Per-stratum NSD+ AUC on the Venuto-expanded dual-anchor cohort A complementary construct-validity test extends beyond the marginal S+ rate disclosure: among NSD-positive PD patients with both anchors classified (observed or Venuto-imputed), does the 21-feat sub-stager perform comparably across anchor configurations? Combining observed `s_positive` with Venuto-imputed values at the threshold $p \geq 0.5$ yields a 4-stratum partition of the NSD-positive cohort: **S+D+ canonical dual-anchor** ($n=271$, of which 236 are Venuto-imputed), **S+D− S-only** ($n=39$), **S−D+ D-only** ($n=52$), and **S−D−** ($n=0$ within NSD+ by NSD-ISS construction). On the canonical S+D+ stratum the 21-feat NSD+ sub-stager achieves macro-OvR AUC **0.912** [95% bootstrap CI 0.886, 0.939], statistically indistinguishable from the full-cohort 5-fold pooled AUC of 0.906 ($\Delta=+0.006$). The discordant strata S+D− and S−D+ are too small (and stage-homogeneous) within the NSD+ cohort for within-stratum AUC evaluation, but their existence is itself the disclosure: 39 NSD+ patients with Venuto-implied S+ but observed D−, and 52 with Venuto-implied S− but observed D+. The Venuto expansion also enables a smoking-gun construct-validity rate: among 779 NSD+-labeled patients, 22 (2.8%) are Venuto-imputed S− at $p < 0.5$, plus 30 with observed-SAA-negative status, giving a combined ~6.7% of NSD+ labels assigned via D-anchor or clinical-signs criteria despite Venuto/observed S−. Per Simuni 2024, single-anchor (D+ with clinical signs) staging is permitted, so this 6.7% is not necessarily a labeling error—but it bounds the dual-anchor-discordant population at clinically tractable size and makes the empirical case that PPMI’s NSD-ISS labels are not D-anchor-only artifacts. A separate disclosure: the Venuto-expansion identifies 431 PPMI patients (mostly Stage 0) with Venuto-implied S+ but D−, who would not be NSD+-classified under the current decision path. These are likely early-prodromal hyposmic patients on the NSD trajectory whose DaT-SBR has not yet crossed threshold; broader SAA screening would re-classify them. Full per-stratum results at `outputs/paper1_r2_responses/q_venuto_4stratum_phase1.json`.

A complementary binary-target stratified analysis (Stage-A NSD+ vs Stage-0 detector, Phase 2) on the same 969-patient Venuto-classified cohort yields a striking corroboration: three of the four anchor strata are *pure-class within target binary by NSD-ISS construction* (D+ implies Stage ≥ 1 , hence NSD+; S−D− implies Stage 0). Only the S+D− stratum mixes 39 NSD+ patients (with elevated clinical signs raising stage despite D−) with 392 Stage-0 patients ($n=431$ total). On this stratum, where D-anchor information is by construction absent, the 21-feat binary classifier’s AUC collapses to 0.490 [bootstrap 95% CI 0.393, 0.592] and the 12-feat clinical-only collapses to 0.432—both at chance level. This is direct empirical evidence that the binary detector discriminates primarily through the D-anchor signal: when D− is held constant, S-anchor + clinical features alone do not separate NSD+ from Stage 0, even with caudate SBR available. The finding is fully consistent with—and quantifies—the paper’s claim that DaT-SPECT is load-bearing for binary detection ($\Delta=-17.6$ pp under DaT removal, the Results (Feature Ablation)); the strict asymmetry between binary (D-dependent) and NSD+ sub-staging (D-independent) is now also visible at the anchor-stratum level (Phase 2 results at `outputs/paper1_r2_responses/q_venuto_4stratum_phase2.json`).

A related feature-engineering limitation concerns the MDS-UPDRS Part III motor score. We exclude the total score NP3TOT from the feature set because it enters the Simuni 2024 clinical staging thresholds directly, but we retain the four UPDRS-III subscales (tremor, rigidity, bradykinesia, axial). Because these

four subscales sum approximately to the excluded total score, the exclusion of NP3TOT is a cosmetic rather than strict information-theoretic decoupling. A sensitivity ablation aggregating the four subscales into a single sum and using it in place of the excluded NP3TOT changed binary CatBoost AUC by <0.01 , confirming that the motor-information leak is bounded; the substantive protection against circularity is provided by excluding the staging-threshold decision boundary itself (the binary “above vs. below the Simuni cutoff” variable), not by dropping the underlying motor-examination information.

12-feature clinical-only NSD+ sub-staging — rule-anchor robustness A potential concern with the 12-feature clinical-only sub-stager (the Results (Feature Ablation), Table 5) is that it retains UPDRS2_TOTAL (ADL) and MOCA_TOTAL—variables that Simuni 2024 uses as clinical staging-threshold anchors—which could yield tautological reproduction of the staging rule. To test this directly, we re-fit CatBoost on three rule-anchor-elided variants of the 12-feat NSD+ sub-stager (5-fold stratified CV, $n=779$ NSD-positive PD patients, fold-local median imputation, 1,000-sample bootstrap CIs): (a) drop UPDRS2_TOTAL alone (11 features); (b) drop UPDRS2_TOTAL and MOCA_TOTAL (10 features); (c) drop UPDRS1_TOTAL, UPDRS2_TOTAL and MOCA_TOTAL (9 features). Macro-OvR AUCs are 0.907 (full 12-feat), 0.900 ($\Delta=-0.007$, 95% CI $[-0.016, -0.000]$), 0.901 ($\Delta=-0.006$, $[-0.014, +0.000]$), and 0.898 ($\Delta=-0.009$, $[-0.020, -0.001]$) respectively—all three rule-anchor exclusions fall well within the pre-registered tolerance of $|\Delta\text{AUC}| < 0.03$, with verdict NO_TAUTOLOGY. The clinical-only NSD+ sub-stager is therefore not recovering the Simuni rule from UPDRS-II/MoCA; the residual predictive signal comes from non-staging variables (UPDRS-III subscales, ESS, RBD, demographics)—consistent with the TreeSHAP attributions in the Results (Feature Ablation) (R6-Q5) showing the top-3 features as UPDRS-III bradykinesia, axial, and rigidity. Full results at `outputs/paper1_r2_responses/q_gap_3.12feat_rule_anchor_ablation.json`.

Missing data presents an additional challenge for external deployment: median imputation with PPMI training-set values is unreliable when entire feature domains are absent, as was the case for HBS (5 of 12 common features missing, reported in Supplementary the Supporting Information). All predictions in this study are cross-sectional baseline snapshots and do not capture the temporal dynamics of disease progression; extending the present benchmark to longitudinal stage-transition prediction is left as future work.

Methodological extensions deferred to future work (R7-Q1, Q2, Q5, Q6) Four reviewer-identified methodological refinements are acknowledged as substantive additions deferred to a dedicated follow-up paper or to the camera-ready revision: (i) *Unified missingness handling* (R7-Q1)—missing-indicator modelling, inverse-probability weighting, or multiple imputation strategies that would unify the internal vs. external feature-set comparability mismatch documented in §III.D; this requires careful causality-sensitive design and is non-trivial under the current $>80\%$ MoCA/UPDRS-IV missingness regime. (ii) *BBSE label-shift estimator* (R7-Q2)—the Lipton *et al.* (2018) Black-Box Shift Estimator is the natural complement to the R3-Q4 Saerens 2002 EM and R6-Q2 Mondrian CP analyses, particularly for unlabelled-target deployments where Mondrian recalibration is not feasible. (iii) *Cascade decision-curve analysis* (R7-Q5)—the R5-Q4 single-stage DCA + R5-Q10 cascade operating-point sweep partially address this, but cascade-specific net-benefit curves under realistic prevalence + referral-budget constraints would directly support clinical-utility claims. (iv) *Alternative clinical-only subset sensitivity* (R7-Q6)—the R5-Q3 striatal-free upper bound establishes the most aggressive removal verdict (NSD+ invariant), but alternative subset definitions (e.g., excluding UPDRS-III subscales jointly with imaging) would map the full transportability–accuracy frontier. We treat all four as future work rather than within-scope additions to keep the present paper focused on the strict-circularity benchmark + uncertainty-aware deployment narrative.

Future external-cohort plans (R7-Q10) The current external evaluation is restricted to BioFIND ($n=103$ NSD-staged via Russo 2025 [32]), with PDBP ($n=893$) and HBS ($n=649$) reported as prediction-only because they lack NSD-ISS ground truth. Two near-term cohort additions are planned to strengthen multiclass external transportability: (i) the Lewy Body Disease cohort (LBD) and the Latino Cohort Study (LCC), both with longitudinal DaT-SPECT—download infrastructure is in place but NSD-ISS staging adaptation has not yet been executed; (ii) the DeNoPa cohort (Mollenhauer *et al.* [26]), pending direct PI collaboration. Additional candidates with treatment-effect or trial designs (e.g. SURE-PD3) require

independent assessment of NSD-ISS staging feasibility under their respective DUAs before they can be incorporated. The aspirational goal of expanding NSD-ISS-labelled external evaluation beyond the $n=103$ BioFIND cohort is acknowledged as a critical pre-deployment requirement; the path forward is concrete but the timing depends on DUA + collaboration approval cycles outside the scope of this paper.

Future Directions

Three extensions of this work merit future investigation. First, stage-conditioned graph-informed imputation would leverage the observation that missingness patterns differ systematically between NSD-ISS stages; imputation models that account for stage-specific patterns may reduce bias and improve downstream prediction, particularly for external cohorts with incomplete features. Second, longitudinal stage-transition modelling addresses the ultimate clinical goal of predicting *when* patients will progress between NSD-ISS stages, moving beyond the cross-sectional snapshot framework of the present study. Third, the healthy-control confound first raised by earlier drafts has been incorporated into the present paper (the Results (Training-Label Confound)) rather than deferred, in response to the Espay *et al.* [15] medication-status critique; a prospective PD-only external cohort is the appropriate next step for statistically powered confirmation.

Conclusion

We present the first comprehensive benchmark for NSD-ISS biological-stage prediction, with four target formulations, cross-conformal uncertainty quantification, five pre-registered confounder analyses (age, sex, enrollment wave, DaT-SPECT protocol, site-LOSO with honest failure disclosure), and a two-stage deployment architecture (Stage-A imaging-anchored detection, Stage-B clinical-only sub-staging). Five tabular-SOTA methods on the 21-feature strict-circularity primary—CatBoost (default) and CatBoost / LightGBM under nested 5×3 CV HPO, TabPFN v2 foundation-model cloud inference, and AutoGluon 1.5 full-pool weighted ensembling—are statistically equivalent at the $\varepsilon=0.02$ practical-equivalence threshold (paired-bootstrap TOST: 34 of 40 pairs equivalent), empirically validating recent convergence predictions at $n \approx 2,000$ clinical cohorts and refining the earlier “trees dominate” narrative. Biological markers are load-bearing for binary staging—removing all DaT-SPECT features drops binary AUC by -17.6 pp ($\Delta = -0.176$, $p < 0.0001$), quantifying the imaging contribution—while clinical features alone suffice for NSD-positive sub-staging (AUC 0.899 vs. 0.908 with imaging, $\Delta = +0.008$, $p = 0.21$), supporting a two-stage deployment pattern at resource-limited sites. External conformal coverage on BioFIND holds for binary (0.915) but collapses for three-class (0.499) and partially retains for NSD+ (0.707), and external calibration degrades $10\times$ relative to internal PPMI; deployment at a new site therefore requires local-cohort recalibration before probabilities can be reported. The PPMI-to-BioFIND domain-shift finding on the binary target is corrected, rather than deferred to future work, by a pre-specified PD-only retraining experiment (the Results (Training-Label Confound)) that directly addresses the Espay *et al.* [15] critique of healthy-control contamination in NSD-ISS reference classes.

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